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Neural feedback facilitates rough-to-fine information retrieval

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ABSTRACT

Categorical relationships between objects are encoded as overlapped neural representations in the brain, where the more similar the objects are, the larger the correlations between their evoked neuronal responses. These representation correlations, however, inevitably incur interference when memories are retrieved. Here, we propose that neural feedback, which is widely observed in the brain but whose function remains largely unknown, contributes to disentangle neural correlations to improve information retrieval. We study a hierarchical neural network storing the hierarchical categorical information of objects, and information retrieval goes from rough-to-fine, aided by the push–pull neural feedback. We elucidate that the push and the pull components of the feedback suppress the interferences due to the representation correlations between objects from different and the same categories, respectively. Our model reproduces the push–pull phenomenon observed in neural data and sheds light on our understanding of the role of feedback in neural information processing.

1. Introduction

Information processing in the brain exhibits a feedforward layered structure, which enables the neural system to extract object features from simple to complex, layer by layer, and this has motivated the development of deep neural networks (DNNs) widely used in AI applications (LeCun, Bengio, & Hinton, 2015). However, the experimental data has revealed that in addition to feedforward connections, there exist abundant feedback connections in the neural pathways, whose number is even larger than that of the former (Sillito, Cudeiro, & Jones, 2006). The importance of neural feedback has been widely recognized in the field, and several theories were proposed to speculate its potential computational functions. For instance, the theory of analysis-by-synthesis proposes that neural feedback, in coordination with the feedforward processing, enables the neural system to recognize an object in an interactive manner (Lee & Mumford, 2003), where the feedforward path extracts object features from external inputs, and the feedback path generates hypotheses about the object identity; the interplay between two pathways accomplishes the recognition task. The predictive coding theory has the similar idea, which assumes that the feedback path predicts the output of the feedforward one (Rao & Ballard, 1999). Although these theories are inspirational, we still lack a clear picture elucidating the detailed computations performed by neural feedback. These include, for instances, what type of object information is conveyed by neural feedback from higher to lower layers, and how this information interacts with the feedforward information to facilitate object recognition. Answering these questions not only addresses neuroscience concerns, but also inspires the further development of DNNs.

Our study on neural feedback is inspired by an experimental finding which unveils a salient feature of neural feedback in the visual system, called the push–pull phenomenon (Chen, Wang, Liang, & Li, 2017; Chen et al., 2014; Gilad, Meirovithz, & Slovin, 2013). Fig. 1A presents the population activity of neurons in the primary visual cortex (V1) when a monkey was carrying out a contour recognition task (Gilad et al., 2013). It shows that the neural population activity increased at the early phase after the onset of the stimulus, displaying the push phenomenon; and it decreased at the late phase, displaying the pull phenomenon. In the push phase, the contributions of the external feedforward input and feedback were mixed, but causality analysis revealed that a feedback component from higher cortex to V1 indeed

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exists (Chen et al., 2017). In the pull phase, multi-unit recording confirmed that there was a strong negative feedback from higher cortex (Chen et al., 2014). Overall, the neural feedback during object recognition exhibits the push–pull characteristic, being first push and then pull. Moreover, when no contour information (i.e., a random background image) was presented to the monkey, the pull phenomenon was weakened significantly.

Why does the neural feedback exhibit the push–pull characteristic? To address this question, let us first review two fundamental issues concerning object recognition in the brain. The first issue concerns the rough-to-fine procedure, namely, the brain first perceives the higher categorical information of an object before recognizing its lower categorical information. This phenomenon has been widely observed in experiments. For instance, Sugase, Yamane, Ueno, and Kawano (1999) found that the higher categorical information of a face, such as being monkey or human, was processed earlier than the detailed information about the expression of the face, such as being happy or sad, in the temporal cortex. It is supposed that the subcortical pathway and/or the dorsal pathway mediate the rapid recognition of the higher categorical information of an image (see more in Discussion). Given that neural object recognition goes from rough to fine and that higher categorical information can enhance the retrieval of lower categorical information, it is reasonable to assume that the neural feedback conveys the higher categorical information of an object from higher to lower layers.

The second issue concerns a critical computational challenge for object recognition in the brain. Object recognition, in essence, is to discriminate the categories of objects based on their similarity/dissimilarity either at the image level or in the semantic sense. The categorizations of objects are organized hierarchically, in the sense that objects belonging to the same category are more similar than those belonging to different ones. A large volume of experimental studies has revealed that the brain encodes the similarities between objects using overlapping neural representations, in the sense that the higher the similarity between the objects, the larger the correlation between their evoked neuronal responses (Carlson, Simmons, Kriegeskorte, & Slevc, 2014; Connolly et al., 2012; Edelman, 1998; Kriegeskorte & Kievit, 2013; Kriegeskorte et al., 2008; Mur et al., 2013; Yamins et al., 2014). For instance, it was found in monkeys that neuronal responses in V4 and IT for objects belonging to the same category have larger correlations than those belonging to different categories (Yamins et al., 2014). However, it is also a well-known fact that a biologically plausible recurrent neural network has difficulty to retrieve correlated neural representations (Hertz, Krogh, & Palmer, 2018). In the classic Hopfield model, it has been shown that once memory patterns are slightly correlated, the retrieval performance of the network is deteriorated dramatically (Hertz et al., 2018). Notably, this correlation interference is not intrinsic to biological neural networks, as it is due to the fact that neural circuits utilize neuronal connections for both information encoding and information retrieving, the so-called in-memory computing (Xia & Yang, 2019). Given that neural correlations are essential to encode the categorical relationships between objects, it prompts a question as how the neural system avoids correlation interference in order to retrieve information reliably.

In this paper, we argue that a computational role of neural feedback is to convey the higher categorical information of an object extracted at higher layers to lower layers, which enhances the retrieval of the lower categorical information of the object. To achieve this goal, the neural feedback needs to have push and pull components, which account for suppressing the interferences due to neural correlations from different and the same categories, respectively. To demonstrate this idea, we study a hierarchical neural network which stores the hierarchical categorical information of objects, as illustrated in Fig. 1C and B, respectively. We investigate the optimal forms of the push and pull feedback components, elucidate their roles on disentangling neural correlations, and demonstrate that the push–pull feedback improves the network retrieval performance significantly. Our model reproduces the push–pull phenomenon observed in neural data and sheds light on our understanding of the functions of feedback in neural information processing. The preliminary results of this study were reported in Liu et al. (2019).

2. A network model for hierarchical information retrieval

To get insight into understanding the role of neural feedback, we consider a hierarchical neural network which stores hierarchical memory patterns, and memory retrieval is described by the Hopfield model. This network model is simple enough for us to pursue the theoretical analysis, but meanwhile captures the nature of information retrieval in a neural network.

The network we consider consists of three layers (Fig. 1C), with each layer having the same number of neurons. Denote the state of neuron i in layer l at time t as $x_i^l(t)$ for $i = 1, \ldots, N$ and $l = 1, 2, 3$, which takes the values of ± 1, the symmetrical recurrent connections from neuron j to i in layer l as W_{ij}^l, the feedforward connections from neuron j of layer l to neuron i in layer $l+1$ as $W_{ij}^{l+1,l}$ and the feedback connections from neuron j of layer $l+1$ to neuron i of layer l as $W_{ij}^{l,l+1}$.

The input received by neuron i in layer l is given by (only the result for layer 1 is shown, and the results for other layers are similar),

$$h_i^1(t) = \sum_j W_{ij}^1 x_j^1(t) + \sum_j W_{ij}^{1,2} x_j^2(t) + I_i^{ext,1}, \tag{1}$$

which includes the recurrent current from neurons in the same layer, the feedback current from neurons in the higher layer, and the external input.

The dynamics of a neuron follows the Hopfield model (Hopfield, 1982), which is written as

$$x_i^l(t+1) = \text{sign}\left[h_i^l(t)\right], \tag{2}$$

where the function $\text{sign}(h) = 1$ for $h > 0$ and $\text{sign}(h) = -1$ otherwise.

We generate synthetic patterns with hierarchical correlations to study memory retrieval theoretically (Parga & Virasoro, 1986). Three generations, from child, to parent, and to grandparent, are considered (Fig. 1B). Denote $\{\xi^\alpha\}$ for $\alpha = 1, \ldots, P_\alpha$ the grandparent patterns, $\{\xi^{\alpha,\beta}\}$ for $\beta = 1, \ldots, P_\beta$ the patterns of parents β who are descendants of grandparent α, and $\{\xi^{\alpha,\beta,\gamma}\}$ for $\gamma = 1, \ldots, P_\gamma$ the patterns of children γ who are descendants of parent (α, β), where P_α, P_β, and P_γ are the number of grandparent patterns, the number of parent patterns belonging to the same grandparent, and the number of child patterns belonging to the same parent, respectively. The details of generating these hierarchical memory patterns are described by Eqs. (A.1)–(A.3) in Appendix A. Overall, these stochastically generated patterns satisfy the hierarchical categorical relationships, in the sense that patterns in the same group have stronger correlation than those belonging to different groups. For example, the correlation between sibling patterns belonging to the same parent is given by $\sum_i \xi^{\alpha,\beta,\gamma} \xi^{\alpha,\beta,\gamma'}/N = b_1^2$, where $0 < b_1 < 1$; the correlation between cousin patterns belonging to different parents but sharing the same grandparent is given by $\sum_i \xi^{\alpha,\beta,\gamma} \xi^{\alpha,\beta',\gamma'}/N = b_1^2 b_2^2$, where $0 < b_2 < 1$ is a parameter specifying the correlation between the parents from the same grandparent; and the correlation between two child patterns belonging to different grandparents is given by $\sum_i \xi^{\alpha,\beta,\gamma} \xi^{\alpha',\beta',\gamma'}/N = 0$. They satisfy the hierarchical relationship, i.e., $b_1^2 > b_1^2 b_2^2 > 0$. Other correlation relationships can be obtained similarly (see Appendix A).

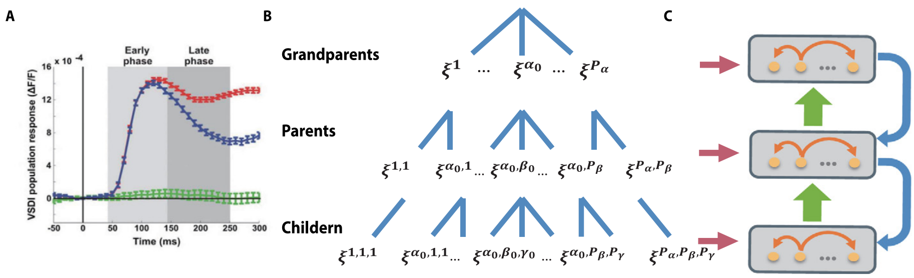

Fig. 1. The push–pull phenomenon and the model structure. (A) The push–pull phenomenon of neuronal responses observed experimentally. Adapted from Gilad et al. (2013). Data recorded by calcium imaging at V1 of an awake monkey. The visual stimulus is a contour embedded in noise, which is onset at $t = 0$. The blue curve shows the change of neural response over time, which increases at the early phase and decreases at the late phase. The red curve represents the response when a random image was presented to the monkey, and the green curve the condition of no visual stimulus. (B) A branching tree displaying the hierarchical categorical relationships of three-level memory patterns, referred to as the grandparents ξ^{α}, the parents $\xi^{\alpha,\beta}$, and the children $\xi^{\alpha,\beta,\gamma}$, respectively, where α, β and γ are indexes indicating the grandparent, parent and child labels of the patterns. The details of generating these hierarchical memory patterns are described in Appendix A. (C) A three-layer network storing the hierarchical memory patterns in (B). Neurons in the same layer are recurrently connected with each other to function as an associative memory, and neurons between layers are connected with both feedforward and feedback connections.

In the hierarchical network, each layer behaves as an associative memory, and the recurrent connections between neurons in the same layer follow the Hebbian rule, which are given by $W^1_{ij} = \sum_{\alpha,\beta,\gamma} \xi_i^{\alpha,\beta,\gamma}\xi_j^{\alpha,\beta,\gamma}/N$, $W^2_{ij} = \sum_{\alpha,\beta} \xi_i^{\alpha,\beta}\xi_j^{\alpha,\beta}/N$, and $W^3_{ij} = \sum_{\alpha} \xi_i^{\alpha}\xi_j^{\alpha}/N$. The feedforward connections between layers also follow the Hebbian rule, e.g., from layers 1 to 2, $W^{2,1}_{ij} = \sum_{\alpha,\beta,\gamma} \xi_i^{\alpha,\beta}\xi_j^{\alpha,\beta,\gamma}/N$. Their contribution is straightforwardly understandable. For instance, if layer 1 is at the state of the memory pattern $\xi^{\alpha_0,\beta_0,\gamma_0}$, then the feedforward input to layer 2 is given by $\sum_j W^{2,1}_{ij}\xi_j^{\alpha_0,\beta_0,\gamma_0} \approx \xi_i^{\alpha_0,\beta_0}$, which helps to retrieve the parent pattern ξ^{α_0,β_0} at layer 2. The forms of feedback connections are the focus of the present study and will be investigated later.

To quantify the retrieval performance, we define a macroscopic variable $m(t)$, measuring the overlap between the network state $\mathbf{x}^l(t)$, for $l = 1, 2, 3$, and a memory pattern, which is calculated to be (Hertz et al., 2018) (again, only the result for layer 1 is shown),

$$m^{\alpha,\beta,\gamma}(t) = \frac{1}{N}\sum_{i=1}^{N} \xi_i^{\alpha,\beta,\gamma} x_i^1(t), \tag{3}$$

where $-1 < m^{\alpha,\beta,\gamma}(t) < 1$ represents the retrieval accuracy of the memory pattern $\xi^{\alpha,\beta,\gamma}$, and the larger the value of m, the higher the retrieval accuracy.

3. Pattern correlation interfering memory retrieval

To explore the effect of neural feedback, we first study a simple case that the network has only two layers (Fig. 2A). Later, we extend the results to three layers. We consider first there is no feedback (setting $W^{1,2}_{ij} = 0$) and inspect how pattern correlations interfere the memory retrieval of the network.

Let the initial state of layer 1 be a memory pattern $\mathbf{x}^1(0) = \xi^{\alpha_0,\beta_0,\gamma_0}$ and the external input $I_i^{ext,1} = 0$. We explore how pattern correlations disturb the stability of the memory state. After one iteration of Eq. (2), we calculate the retrieval accuracy of layer 1, which is given by (see Appendix B),

$$\begin{aligned} m^{\alpha_0,\beta_0,\gamma_0}(1) &= \frac{1}{N}\sum_{i=1}^{N} \xi_i^{\alpha_0,\beta_0,\gamma_0} x_i^1(1), \\ &= \frac{1}{N}\sum_{i=1}^{N} \xi_i^{\alpha_0,\beta_0,\gamma_0} \operatorname{sign}\left[h_i^1(0)\right], \\ &= \frac{1}{N}\sum_{i=1}^{N} \operatorname{sign}\left[\xi_i^{\alpha_0,\beta_0,\gamma_0} h_i^1(0)\right]. \end{aligned} \tag{4}$$

It shows that the retrieval of a memory pattern is determined by the alignment between the neural input and the memory pattern, which can be further written as (see Appendix B),

$$\xi_i^{\alpha_0,\beta_0,\gamma_0} h_i^1(0) = \xi_i^{\alpha_0,\beta_0,\gamma_0} \sum_j W^1_{ij} x_j^1(0) = 1 + C_i + \widetilde{C}_i. \tag{5}$$

Here, the input received by the neuron is decomposed into the signal and the noise parts, and the latter is further divided into two components, which are given by

$$C_i = b_1^2 \xi_i^{\alpha_0,\beta_0,\gamma_0} \sum_{\gamma\neq\gamma_0} \xi_i^{\alpha_0,\beta_0,\gamma}, \tag{6}$$

$$\widetilde{C}_i = b_1^2 b_2^2 \xi_i^{\alpha_0,\beta_0,\gamma_0} \sum_{\beta\neq\beta_0}\sum_{\gamma} \xi_i^{\alpha_0,\beta,\gamma}. \tag{7}$$

We see that C_i and $\widetilde{C}_i$ represent, respectively, the interferences to memory retrieval due to: (1) the correlation of a pattern with its siblings from the same parent, called the **intra-class interference**; and (2) the correlation of a pattern with its cousins from the same grandparent but different parents, called the **inter-class interference**. A retrieval error occurs when the noise amplitude $C_i + \widetilde{C}_i < -1$.

In the limits of large N, P_γ and P_β, the intra- and inter-class interferences, C_i and $\widetilde{C}_i$, satisfy the distributions, $P(C_i) = \mathcal{N}(E_C, V_C)(1+b_1)/2 + \mathcal{N}(-E_C, V_C)(1-b_1)/2$, $P(\widetilde{C}_i) = \mathcal{N}(E_{\widetilde{C}}, V_{\widetilde{C}})(1+b_1b_2)/2 + \mathcal{N}(-E_{\widetilde{C}}, V_{\widetilde{C}})(1-b_1b_2)/2$, where $\mathcal{N}(E, V)$ represents the normal distribution with mean E and variance V, and $E_C = b_1^3(P_\gamma - 1)$, $E_{\widetilde{C}} = b_1^3 b_2^3 P_\gamma(P_\beta - 1)$, $V_C = b_1^4(P_\gamma - 1)(1 - b_1^2)$, $V_{\widetilde{C}} = b_1^4 b_2^4 P_\gamma(P_\beta - 1)(1-b_1^2)(1-b_2^2)$ (see Appendix B). Example distributions of the intra- and inter-class interferences are displayed in Fig. 2B. Note that the intra-class interference is bimodal. The higher positively-positioned peak corresponds to interference at the neurons where the memory patterns of a child are the same as those of the parent, and the lower negatively-positioned peak corresponds to those where the memory patterns of the child are opposite to those of the parent. In contrast, the inter-class interference embodies a broader range of correlations among the memory patterns, and becomes much more unimodal.

4. Rough-to-fine information retrieval aided by feedback

The above theoretical analysis tells us that even starting from a clean memory state, the pattern correlations can disturb the network state, causing the instability of the memory state (Hertz et al., 2018). The result of Eq. (5) further suggests that to improve memory retrieval, the key of neural feedback is to suppress the

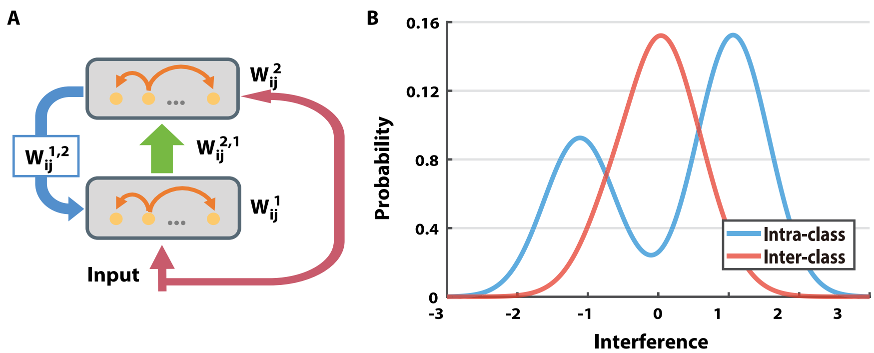

Fig. 2. (A) A two-layer network extracted from Fig. 1(C) for analyzing the effect of neural feedback theoretically. (B) Example distributions of the intra- and inter-class interferences in memory retrieval. The calculation detail is given in Appendix E.

inter- and intra-class interferences due to different sources of pattern correlation.

In reality, the correlation between higher categorical patterns tend to be smaller than that between lower categorical patterns. For example, the similarity between cats and dogs is usually smaller than that between two sub-types of cats or dogs. In our model, this corresponds to the condition $b_1 > b_2$. Since pattern correlation is the main cause of retrieval error in a neural network, this implies that given the same amount of input information (e.g., an ambiguous image of Persian cat), the parent pattern (e.g., cat) can be better retrieved than the child pattern (e.g., Persian cat). Therefore, we consider a rough-to-fine information retrieval procedure, in which the memory patterns in higher layers are first retrieved, whose results are subsequently fed back to lower layers to improve the retrieval of lower memory patterns. This idea is in agreement with the reserve hierarchy theory of perception (Ahissar & Hochstein, 2004) (see more in Discussion).

For convenience of theoretical analysis, we continue to consider the two-layer network and assume that the parent pattern is first perfectly retrieved ($m = 1$), and explore the appropriate forms of push and pull feedbacks which can effectively utilize the parent information to facilitate the retrieval of the child pattern. Later, we carry out simulations to demonstrate that the model works well in general cases.

5. Push feedback suppressing the inter-class interference

We first show that a push feedback of a proper form can suppress the inter-class interference effectively in memory retrieval. Without loss of generality, we consider the case that for a given input, the target child pattern to be retrieved in layer 1 is $\xi^{\alpha_0,\beta_0,\gamma_0}$ and the corresponding parent pattern in layer 2 is ξ^{α_0,β_0}, which has already been retrieved. We propose that the push feedback has the below form,

$$W_{ij}^{1,2} = \frac{1}{NP_\gamma} \sum_{\alpha,\beta,\gamma} \xi_i^{\alpha,\beta,\gamma} \xi_j^{\alpha,\beta}, \tag{8}$$

which follows the standard Hebb learning rule between the parent and its child patterns.

The contribution of this push feedback form is intuitively understandable. Given that the parent pattern ξ^{α_0,β_0} in layer 2 is already retrieved, the push feedback current it generates to neuron i in layer 1 is calculated to be $\sum_j W_{ij}^{1,2}\xi_j^{\alpha_0,\beta_0} \approx \sum_\gamma \xi_i^{\alpha_0,\beta_0,\gamma}/P_\gamma$. Obviously, this push current increases the activities of all sibling patterns belonging to the parent, i.e., those $\xi^{\alpha_0,\beta_0,\gamma}$ for any γ, and it has little influence to cousin patterns from different parents, i.e., those $\xi^{\alpha_0,\beta,\gamma}$ for $\beta \neq \beta_0$. Thus, in effect, the push feedback suppresses the interference due to inter-class noises (see Fig. 3A).

Notably, the major function of the push feedback is not to improve the retrieval accuracy of the target child pattern directly, and may even worsen it in certain cases (see Fig. 3B). Rather, by suppressing the activities of cousin patterns, it helps the subsequent pull feedback to diminish the intra-class interference from sibling patterns (see below).

6. Pull feedback suppressing the intra-class interference

We further show that a pull feedback of an appropriate form can effectively suppress the intra-class interference in memory retrieval. We consider that the pull feedback with the form,

$$W_{ij}^{1,2} = -b_1\delta_{ij}. \tag{9}$$

Given that the parent pattern ξ^{α_0,β_0} in layer 2 is retrieved, the pull feedback received by neuron i in layer 1 is calculated to be $\sum_j W_{ij}^{1,2}\xi_j^{\alpha_0,\beta_0} = -b_1\xi_i^{\alpha_0,\beta_0}$. In the large P_γ limit, the parent pattern is proportional to the mean of its sibling patterns (see Appendix B.2), thus, intuitively the pull feedback is to subtract the mean of all sibling patterns, which effectively highlights the subtle differences between sibling patterns. Referring to Fig. 3C, the positively-positioned peak of the intra-class interference is shifted in the negative direction. More importantly, the negatively-positioned peak is shifted in the positive direction, significantly reducing the probability of a retrieval error caused by the total interference being less than -1. It can be checked that with this pull feedback, sibling patterns become statistically independent, i.e., $\sum_i(\xi_i^{\alpha_0,\beta_0,\gamma} - b_1\xi_i^{\alpha_0,\beta_0})(\xi_i^{\alpha_0,\beta_0,\gamma'} - b_1\xi_i^{\alpha_0,\beta_0}) \approx 0$ (see Appendix B.4), indicating that the pull feedback can reduce the intra-class interference.

After applying the pull feedback, the retrieval accuracy of the target child pattern is calculated to be (see Appendix B.4)

$$m^{\alpha_0,\beta_0,\gamma_0} = \frac{1}{N}\sum_{i=1}^{N} \text{sign}\left[1 + C_i^* + \widetilde{C}_i\right], \tag{10}$$

where $C_i^* \equiv C_i - b_1\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0}$ denotes the new noise term due to intra-class interference after applying the pull feedback (in contrast with Eq. (5)). As shown in Fig. 3C, with the pull feedback, the negative tail of the noise distribution (where retrieval errors occur) is significantly reduced. Fig. 3B presents an example demonstrating scenario that with the pull feedback, a memory state of the network becomes stable and is no longer deteriorated by pattern correlation. We also prove theoretically that if the number of child patterns of a parent is sufficiently large, the pull feedback guarantees the reduction of the retrieval error (see Appendix B.7).

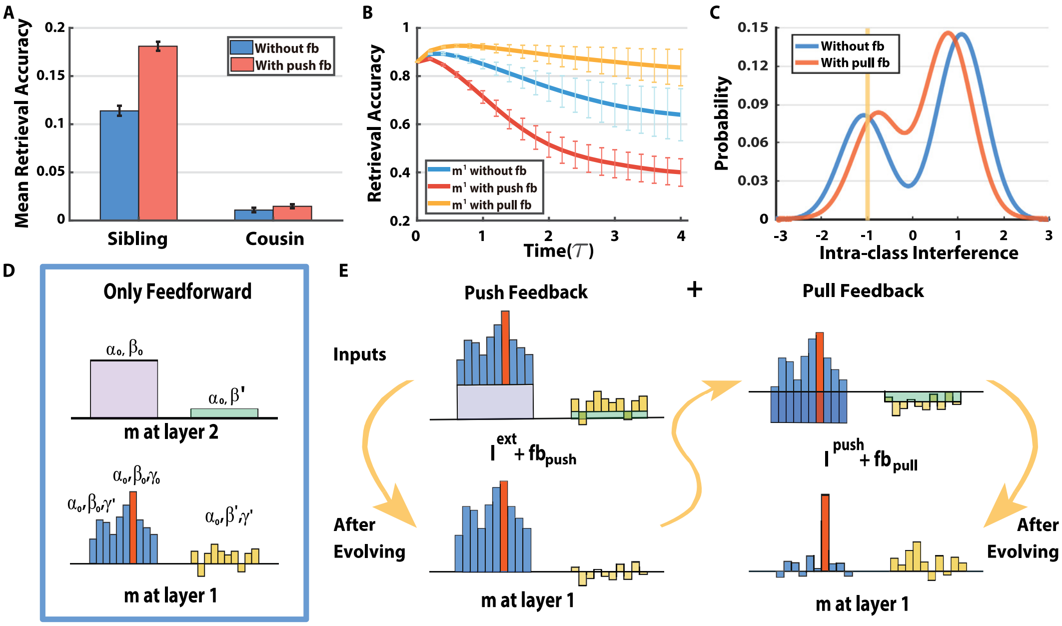

Fig. 3. Illustrating the effects of the push and pull feedbacks. (A) The effect of the push feedback, which increases the mean retrieval accuracy of all sibling patterns and has little influence on cousin patterns. The averaged retrieval accuracies of sibling and cousin patterns are measured by $\sum_\gamma m^{1,1,\gamma}/P_\gamma$ and $\sum_\gamma m^{1,2,\gamma}/P_\gamma$, respectively. (B) The stability of a memory state with no feedback, the push feedback, or the pull feedback, respectively. The initial state of layer 1 is $\xi^{1,1,1}$. Only the pull feedback stabilizes the memory state. (C) The effect of the pull feedback, illustrated by the distributions of the intra-class interference C_i and the new noise term C_i^* after receiving the pull feedback. Retrieval error occurs when the noise value is smaller than -1. The shaded region indicates the improvement of retrial performance after applying the pull feedback. (D–E) Illustrating the effect of push–pull feedback. (D) Without feedback, the parent pattern in layer 2 is retrieved relatively well, but the child pattern in layer 1 is largely confounded with the sibling and cousin patterns. (E). Left panel: the push feedback leverages the total inputs to sibling patterns (top), which increases the retrieval of sibling patterns and suppresses the retrieval of cousins (bottom); Right panel: the pull feedback decorrelates sibling patterns via subtracting their mean (i.e., the parent pattern) (top), which increases the retrieval of the target child pattern (bottom). The simulation details for (A–C) are described in Simulation protocols in Appendix E.

7. Memory retrieval with the push–pull feedback

In summary, the above analyses suggest that to suppress the interference of pattern correlations in memory retrieval, the neural feedback needs to be dynamical, varying from push to pull, so that it can suppress the inter- and intra-class interferences, respectively (see Fig. 3D–E).

To better demonstrate the joint effect of the push–pull feedback, we consider a continuous version of the Hopfield model, so that the network state changes smoothly and the contributions of push and pull feedback can be integrated over time (the discrete Hopfield model still works, but the overall effect is less significant). The network dynamics are given by (Hopfield, 1984),

$$\tau \frac{dh_i^n}{dt} = -h_i^n + \sum_j W_{ij}^n x_j^n + \sum_k W_{ik}^{n,m}(t) x_k^m + I_i^{ext,n}, \tag{11}$$

$$x_i^n = f(h_i^n), \qquad m, n = 1, 2, \tag{12}$$

where h_i^n and x_i^n denote the synaptic input and the firing rate of neuron i in layer n, respectively, and their relationship is given by a sigmoid function, $f(h) = \arctan(8\pi h)/\pi + 1/2$. The parameter τ is the time constant. The recurrent and feedforward connections follow the standard Hebb rule as described above. The feedbacks are slightly modified from Eqs. (8)–(9) to accommodate positive neural activities in the continuous model. They are given by: the push feedback $W_{ik}^{1,2} = a_+ P_\gamma \sum_{\alpha,\beta,\gamma}(\xi_i^{\alpha,\beta,\gamma} - \langle\xi\rangle)(\xi_k^{\alpha,\beta} - \langle\xi\rangle)/N$, with a_+ being a positive number, and the pull feedback $W_{ik}^{1,2} = -a_- b_1 \delta_{ik}$, with a_- being a positive number. $I_i^{ext,n}$ is the external input, which conveys the pattern information. For details of the model, see Appendix C.1.

In the simulation, the push and pull feedbacks are applied sequentially, with each of them lasting in the time order of τ ($\tau \sim$ 10–20 ms), as suggested by the data (Chen et al., 2014). To achieve rough-to-fine information retrieval, we apply the external input simultaneously to layers 1 and 2. This implies that in addition to the feedforward external input to the lowest layer, there exists a shortcut which conveys the external information directly to the higher layer of the network. The biological plausibility of this setting will be discussed later (see Discussion).

7.1. Reproducing the push–pull neural response

We first demonstrate that our model can reproduce the push–pull phenomenon in neural population responses as observed in the experiment (Gilad et al., 2013). Fig. 4A displays the neural activities in a typical example of the memory retrieval process in our model, which shows that the neural population activity at layer 1 first increases and then decreases, displaying the push–pull characteristic. Fig. 4B further confirms that information retrieval goes from rough to fine, in the sense of that the parent pattern is retrieved first, and the retrieval accuracy of the child pattern is improved gradually by the push–pull feedback.

7.2. Evaluating model performances

We systematically evaluated the performances of our model. Fig. 4C displays how the retrieval improvement due to the push–pull feedback (measured by the ratio between the retrieval accuracies with and without feedback) varies with the intra-class correlation amplitude, which shows that our model works very well for a wide range of correlation strengths. We also note that when the correlation strength is too large, the pull feedback may worsen the retrieval performance. Mathematically, it is due to the shut down of the neural activity at layer 1 due to the pull feedback being too strong (see Eq. (9)), but this only happens when $b_1 > 0.4$, i.e., the overlap between the parent and its

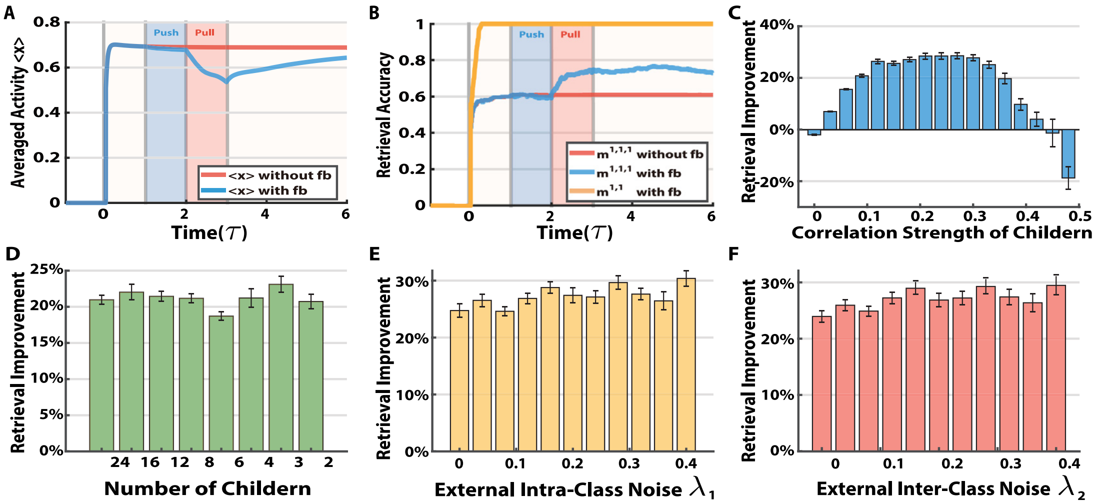

Fig. 4. The model performances with the push–pull feedback. (A) The neural population activity $\langle x \rangle$ in layer 1 as a function of time in a typical trial (blue curve), which exhibits the push–pull phenomenon as observed in the experiment (Gilad et al., 2013). The red curve is the case without feedback. $\langle x \rangle$ is obtained by averaging over all neurons in layer 1. The shaded parts in A and B display the time period of applying the external input (beige, $0-6\tau$), the push feedback (blue, $\tau-2\tau$), and the pull feedback (red, $2\tau-3\tau$), respectively. (B) The retrieval accuracies of the child (blue curve) and the parent patterns (yellow curve) over time in the same trial as in (A) The red curve is the case of no feedback. (C) The retrieval improvement due to the push–pull feedback vs. the intra-class correlation strength (b_1). (D) The network performance vs. the number of child patterns. The total number of memory patterns ($P_\alpha \times P_\beta \times P_\gamma$) is fixed to be 96 and $P_\alpha = 2$. (E) The retrieval improvement due to the push–pull feedback vs. the amplitude of intra-class noise in the external input. (F) The retrieval improvement due to the push–pull feedback vs. the amplitude of inter-class noise in the external input. The simulation details are given in Appendix E.

child patterns is more than $(1+b_1)/2 > 70\%$, which is rare in practice. Fig. 4D displays how the retrieval improvement of the model varies with the number of child patterns (the total number of patterns stored in the network is fixed), which shows that the model works well consistently.

Fig. 4E–F display how the retrieval improvement varies with the amplitude of input noises. We consider two types of input noise, the inter- and intra-class noises, reflecting the ambiguities arisen from that target child pattern being confounded with its cousin or sibling patterns, respectively (see Appendix E). We observe that the push–pull feedback works well for a wide range of noise amplitudes. The results for very large noises are not presented, since in this case, the retrieval accuracies with and without feedback are both very small, and it is meaningless to do the comparison.

8. Structure analysis of the model

8.1. Both push and pull feedbacks are indispensable

We carry out experiments demonstrating that both the push and pull feedbacks are indispensable, in terms of that if only one of them is applied, the network performance is degraded considerably. Fig. 5 compares the network performances with only push, only pull, or the push–pull, feedback, respectively, which shows that the network has the best performance when both the push and pull feedbacks are applied.

It may appear that the pull feedback is completely dominating, while the effect of push feedback is negligible. Actually, this varies with the parameters. We carry out an experiment with different parameters to demonstrate that the push feedback can have a stronger effect than the pull feedback (see Fig. C.8).

8.2. Push–pull, rather than pull–push, is effective

We further confirm the importance of applying the push and pull feedbacks in the right temporal order. Fig. 5C shows that if the pull feedback is applied before the push one, the network performance is degraded. The underlying reason is: the push feedback reduces the activities of cousin patterns (which is done indirectly through increasing the activities of sibling patterns), therefore if the push feedback is applied first, it helps the subsequent pull feedback to reduce the interference of cousin patterns; whereas, if the push feedback is applied after the pull one, it will indiscriminately increase the activities of all sibling patterns and destroys the result of the pull feedback.

9. Performances of a three-layer network

The above study is based on a two-layer network, but the computational principle can be applied straightforwardly to networks having multiple layers. To demonstrate this, we consider a three-layer network as shown in Fig. 6, which stores, from bottom to top, the child, the parent, and the grandparent patterns as given in Appendix A. Compared with the two-layer network in Fig. 2, the external input is sent to layer 3 directly bypassing layer 2, and push–pull feedback connections from layer 3 to 2 are added. Similarly, information retrieval in the network goes from rough-to-fine, i.e., the grandparent pattern in layer 3 is first retrieved, whose result is subsequently utilized to enhance the retrieval of the parent pattern in layer 2, and then the parent information is used to enhance the retrieval of the child pattern in layer 1. The network can exploit multiple rounds of push–pull feedback. In this example, we find that two rounds of push–pull feedback outperforms one round, but the performance does not increase further if more than two rounds are applied. Therefore, we present the results of applying two rounds of push–pull feedback.

Fig. 6B displays the retrieval accuracies of three layers in one example trial, which shows that with the push–pull feedback, the retrieval accuracy of layer 1 is improved significantly compared to the case of no feedback. Fig. 6C presents the statistical results of retrieval improvements of three layers due to the feedback. It is interesting to note that for a wide range of parameter values, the retrieval accuracies of three layers are all improved. This shows that in the presence of feedback, the retrieval accuracies of layers 1/2 are improved, which in return can improve the retrieval accuracy of layers 2/3 via the feedforward connections.

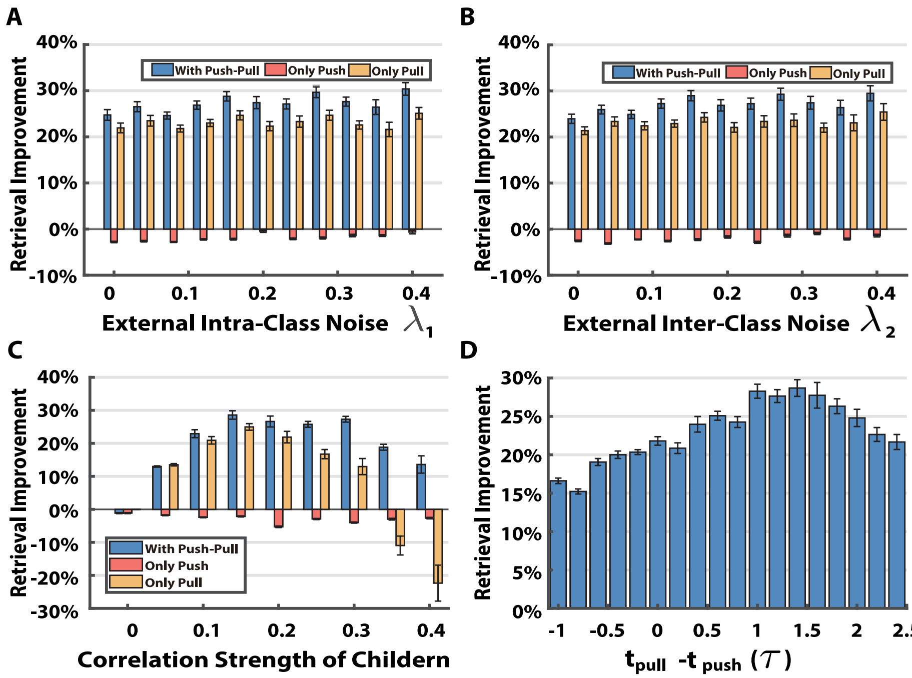

Fig. 5. The necessity of both push and pull feedbacks. (A–C) Comparing the network performances with the push–pull, only the push, and only the pull feedbacks, respectively. (A) The amplitude of inter-class noise in the external input is varied. (B) The amplitude of intra-class noise in the external input is varied. (C) The intra-class correlation strength (b_1) is varied. (D) The network performances with varying moments of applying the push and pull feedbacks. The starting instant of applying the pull feedback ($t_{pull} = 2\tau$, lasting for 1.8τ) is fixed, and the starting instant of applying the push feedback (t_{push}, lasting for 1τ) changes. $t_{pull} - t_{push} > 0$ corresponds to the case of push–pull feedback, and $t_{pull} - t_{push} < 0$ the case of pull–push feedback. For the details of simulation, see Appendix E.

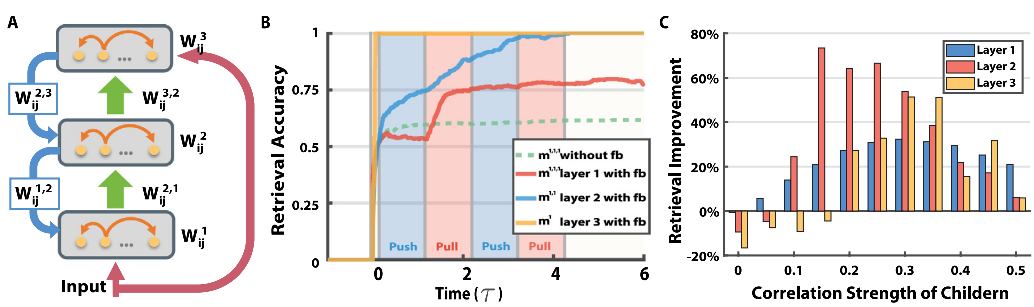

Fig. 6. The performance of the three-layer network. (A) A three-layer network. The external input is sent to layers 1 and 3 simultaneously. (B) The retrieval accuracies of the child (red curve), the parent (blue curve), and the grandparent (yellow curve) over time. The dash green curve is the retrieval accuracy of the child in the case of no feedback. The shaded parts display the time of applying the external input (beige, $0 - 6\tau$), the push feedback for both layer 3 to 2 and 2 to 1 (blue, $0.2\tau - 1.2\tau$; $2.2\tau - 3.2\tau$), and the pull feedback for both layer 3 to 2 and 2 to 1 (red, $1.2\tau - 2.2\tau$; $3.2\tau - 4.2\tau$), respectively. (C) Retrieval improvements with varying intra-class correlation strength b_1. The details of simulation are described in Appendix E.

Note that when the correlation strength is very small ($b_1 < 0.1$), the performance of layer 3 may be degraded. This is understandable, since when the correlation is too small, the categorical relationship between the child and the grandparent patterns become very weak, and the feedforward and feedback interactions no longer guarantee to improve the retrieval performance.

10. Real image experiments

We further test our model in processing real images. As shown in Fig. 7A, the dataset we use consists of $P_\beta = 3$ types of animals, birds, cats, and dogs, corresponding to the parents in our model. For each type of animal, it is further divided into $P_\gamma = 4$ subtypes, corresponding to the children. A total of 1200 images, with $K = 100$ for each sub-type of animals, are chosen from ImageNet. It has been shown that the neural representations generated by a deep neural network (DNN) after trained with ImageNet capture the categorical relationships between objects, in the sense that the overlap between neural representations reflects the closeness of objects in the same category, rather than their similarity in raw images (Yamins et al., 2014). This satisfies the statistical property of hierarchical memory patterns. We therefore pre-process images by a DNN called VGG (Simonyan & Zisserman, 2014) and use the neural representations generated by it (i.e., the neural activities before the read-out layer) to construct the memory patterns. The details of pre-processing are described in Appendix D. Fig. 7B shows the correlations between the memory patterns generated by VGG, which exhibit a hierarchical relationship, i.e., the intra-class correlations are larger than the inter-class correlations.

We present each image to the network and measure its retrieval accuracy by calculating the overlap between the network response and the memory pattern corresponding to the image (i.e., $m^{\beta,\gamma}$, according to Eq. (4)). Fig. 7B shows a typical example of the retrieval process. We see that the retrieval accuracy of layer 1 keeps increasing when the push and pull feedbacks are applied sequentially, and the result is significantly improved compared

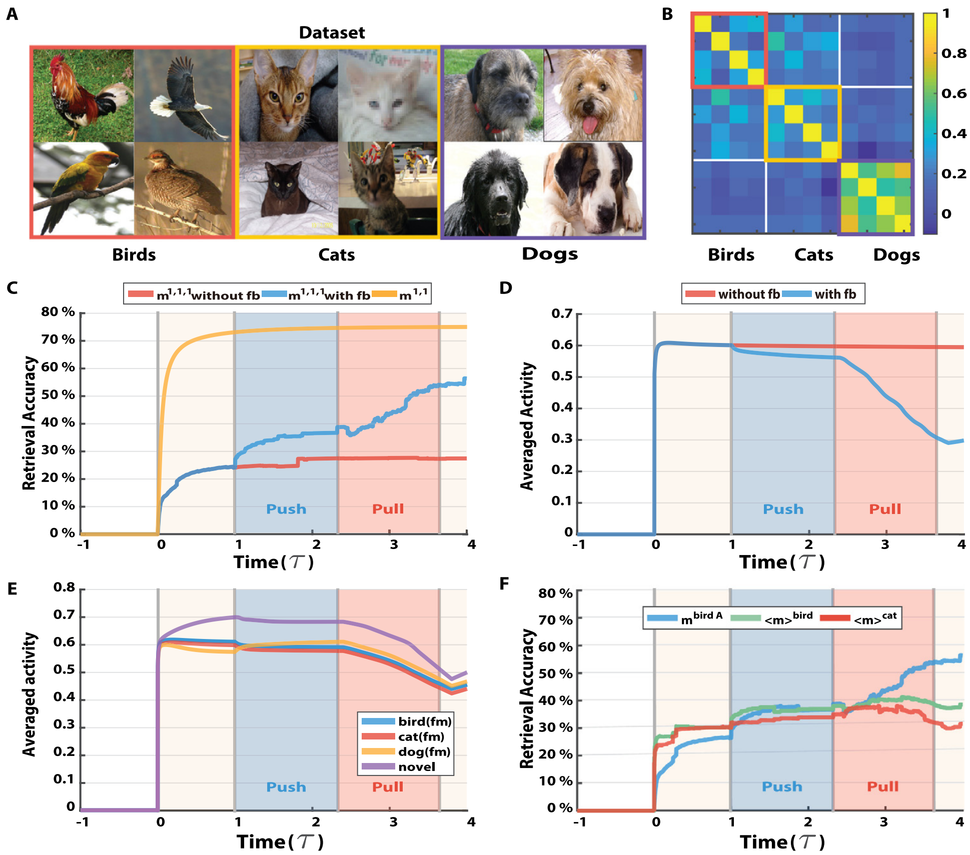

Fig. 7. The model performances with real images. The two-layer network model as shown in (Fig. 2A) is used. (A) The dataset. Example images, one for each sub-type of birds, cats or dogs. Birds: $1-4$; Cats: $5-8$; Dogs: $9-12$. (B) The correlations between the child patterns after pre-processing by VGG. (C) Retrieval accuracies of the child (bird A, cock, blue curve) and parent (bird, yellow curve) patterns as functions of time in a typical trial. The red curve is the case without feedback. (D) The neural population activity $\langle x \rangle$ in layer1 as a function of time in a typical trial (blue curve), which exhibits the push–pull phenomenon as observed in the experiment (Gilad et al., 2013). The blue curve is the case without feedback. (E) The blue, red, yellow and violet curves represent the mean neural population activities in layer 1 with respect to familiar (bird, cat, and dog) and novel (car) images, respectively. (F) Different effects of the push and pull feedback in the example trial as in (C). The blue, green, and red curves represent, respectively, the retrieval accuracies of the target child (bird A, cock), the mean of siblings (other sub-types of bird), and other child patterns (all sub-types of cat) in layer 1. The shaded parts of (C–F) display the time of applying the external input (beige, $0-4\tau$), the push feedback (blue, $1\tau-2.4\tau$), and the pull feedback (red, $2.4\tau-3.8\tau$), respectively. The simulation details are given in Appendix E.

to the case without feedback. Over 1200 images, the averaged improvement is 171.88% (measured at the moment when the pull feedback stops).

10.1. Reproducing the push–pull neural response

We check the dynamical properties of neuronal responses in the model. Fig. 7D displays the time course of neural activities in a typical example of the memory retrieving process, which shows that the neural activity at layer 1 first increases and then decreases, displaying the push–pull phenomenon.

In the monkey experiment, it was also observed that if an unfamiliar image (which did not contain the object the monkey was trained to recognize) was presented, the push–pull characteristic of neural activity was weakened significantly (see the red line in Fig. 1A). To model this phenomenon, we present a set of novel images (they are not stored in our associative memory model), cars, to the network model, and measure the corresponding neural responses. As shown in Fig. 7E, compared to the familiar images that have been stored in the network (birds, cats, and dogs), the push–pull characteristic of neural responses with respect to the novel images is weakened considerably, agreeing with the experimental finding (for the detail of the experiment, see in Appendix D).

10.2. Push–pull feedback disentangling pattern correlations

To demonstrate the different roles of the push and pull feedbacks in memory retrieval, we separately measured the retrieval accuracies of the target child pattern, its siblings, and its cousins, respectively, during the retrieving process. As shown in Fig. 7F, we observe that: (1) at the early phase of applying the push feedback, both the retrieval accuracies of the target child pattern and its siblings increase, whereas the retrieval accuracy of cousins remains almost unchanged, confirming that the push feedback indeed has the effect of suppressing the inter-class interference as analyzed in theory; (2) at the later phase of applying the pull feedback, the retrieval accuracy of the target child pattern experiences another significant increase, much larger than the that for other child patterns, which confirms that the pull feedback has the effect of suppressing the intra-class interference.

11. Discussion

In the present study, we have investigated the computational role of feedback connections in neural information processing, and demonstrated that the push–pull feedback facilitates rough-to-fine information retrieval. We studied a multilayer network which stores the hierarchical categorical information of objects. Upon receiving an external input, the higher categorical information of the object is first retrieved, whose result is subsequently utilized to enhance the retrieval of the lower categorical information of the object via the push–pull feedback, where the push and pull feedback components suppress the inter- and intra-class interferences, respectively. In such a way, the highly correlated neural representations are disentangled, and the retrieval performance of the network is significantly improved. Our study has some far-reaching implications to neural information processing.

In our model, we consider that the neural feedback has the push–pull characteristic, which reproduces the push–pull phenomenon as observed in the experiments (Chen et al., 2014; Gilad et al., 2013; Gilbert & Li, 2013). The time period of applying push–pull feedback is about 100–250 ms, that is consistent with recent experimental studies. By perturbing the propagation of neuronal activity between areas V1 and higher layers by microstimulation or transcranial magnetic stimulation (TMS), it unveils that there exists feedback from higher layers to V1 after 100 ms (Klink, Dagnino, Gariel-Mathis, & Roelfsema, 2017; Li, Wang, & Li, 2019). Notably, the neural system has resources to implement such a dynamical feedback process. For example, the push component of the feedback may be realized via the direct excitatory synapses from higher to lower layers, and its termination can be automatically controlled by short-term synaptic depression (Zucker & Regehr, 2002); while, the pull component of the feedback may go through a separate path mediated by inhibitory interneurons between the layers, which is naturally delayed compared to the direct excitatory path (Kandel, Schwartz, & Jessell, 2000). Our study revealed that the pull feedback contributes to suppressing the intra-class interference, and this result also appears to agree with the experimental finding that different levels of GABA release were associated with different object recognition tasks. Combining MR Spectroscopy measurement with fMRI, Frangou et al. found that the decreasing of GABA was associated with the task of detecting a signal from a noise background; whereas, the increment of GABA was associated with the task of discriminating two patterns of subtle differences (Frangou, Correia, & Kourtzi, 2018). Considering that discriminating similar patterns is much harder than detecting a signal from background and it needs to suppress the intra-class interference, our model predicts that the pull feedback should be involved in the discrimination task, which leads to the increased GABA release, agreeing with the experimental finding.

In our model, we consider that neural information retrieval goes from rough to fine, which is a paradigm that has been widely observed in the experimental data (Chen, 1982, 2005). For example, psychophysical studies showed that the brain extracts first the global (e.g., topological), rather than the local (e.g., Euclidean), geometrical properties of objects (Chen, 1982). The neural pathway for this global information extraction is supposed to be mediated by M cells in the retina, followed by the subcortical and/or the dorsal pathways (Liu, Wang, Zhou, Ding, & Luo, 2017). In our work, we mimic the effect of this short-cut route by considering that a copy of the external input is sent directly to the highest layer of the network, which triggers the retrieval of the highest categorical information of the object first.

On the computational aspect, our model resolves a fundamental challenge for object recognition in the brain. On one hand, neural correlations are essential to encode the categorical relationships (similarities) between objects; on the other hand, they are harmful as they interfere the retrieval of memory patterns. This dilemma is intrinsic to a neural network, as it uses neuronal connections for both information storage and information retrieval, the so-called in-memory computing. How to overcome the correlation interference in memory retrieval is a long-standing challenge in the field. In the classic Hopfield model, when partial or ambiguous information of a memorized object is presented to the network, the memory pattern can be retrieved reliably, achieving associative memory (Amari, 1972; Hopfield, 1982). However, when a member pattern of a family of correlated patterns is presented, the retrieval process is interfered by the correlations with other member patterns of the same family, resulting in mixed retrieved patterns. In fact, considering how information is encoded in synaptic connections through correlated spikes of neurons, direct Hebbian plasticity cannot improve the situation, as correlations among neurons are inherent among member patterns of the same family. This causes the network dynamics to degenerate into a small number of merged attractors (Hertz et al., 2018). To rescue this fatal flaw, modifications of Hebbian plasticity rules were proposed, including those that take into account the hierarchical organization of the patterns (Cortes, Krogh, & Hertz, 1987; Engel, 1990; Parga & Virasoro, 1986; Perez-Vicente, 1989; Tsodyks, 1990), or based on the popularity of neurons (Kropff & Treves, 2007). In the context of learning correlated patterns in continuous attractor neural networks (CANNs), a novelty-facilitated plasticity rule enables the storage of continuously morphed patterns as a sub-manifold with a flat energy landscape in the state space (see Zou et al. (2017) and references therein). However, information retrieval in a CANN is still unreliable, as its dynamics is susceptible to input fluctuations (see Wu, Wong, Fung, Mi, and Zhang (2016) and references therein). In this study, we propose a novel biologically plausible strategy to resolve the correlation interference problem. We consider that the categorical representations of objects are organized hierarchically with reducing correlations along the hierarchy, and the neural system stores these hierarchical memory patterns in a hierarchical network accordingly; upon receiving the input information of an object, the representation of the object at the highest layer is first retrieved (since memory patterns at the highest layer have the minimum correlation), which subsequently initiates the push–pull feedback to improve memory retrieval at the descending layer, and this process goes on till to the lowest layer. By this, the highly correlated memory patterns can be retrieved properly by the neural system.

11.1. Relationship to previous works

To our knowledge, our work is the first one that explores the computational role of push–pull feedback in hierarchical memory retrieval. Hierarchical memory retrieval was studied previously, but those works considered only a single layer network, in which all hierarchical memory patterns are stored jointly via Hebbian learning by the same set of neuronal connections and no feedback is involved (Cortes et al., 1987; Engel, 1990; Matsumoto, Okada, Sugase-Miyamoto, & Yamane, 2005; Parga & Virasoro, 1986; Perez-Vicente, 1989; Tsodyks, 1990). Matsumoto et al. (2005) showed that such type of a network can also realize a certain level of rough-to-fine information processing by having the fast excitatory interactions among excitatory neurons first retrieving the mixed memory states (the rough information), and the relatively slow global inhibition from inhibitory neurons further retrieving the fine information. This computational mechanism is very different from the one proposed in our model. It will be interesting to explore in future works which mechanism is biologically more plausible or they are both plausible but applied in different brain regions.

Hierarchical Hopfield models were also studied previously, but those works have considered neither hierarchical memory patterns nor feedback (Amari & Maginu, 1988; Okada, 2006). An exception is an early work on multiplayer model that used feedback information to disentangle the correlated patterns, in which information from an upper layer is fed back to the lower layer to gate the synaptic couplings, which increases the storage capacity of the network (Sourlas, 1988). However, the feedback form in this study was static, which is different from the push–pull feedback we consider, and this study also did not consider the dynamical nature of the memory retrieval process.

Notably, the conventional studies on Hopfield-type of models normally focus on analyzing the memory capacities of the models, where the memory capacity is defined as the number of stationary states that can be retained reliably by the network after removing external cues. Here, our focus is on the dynamical retrieval process of the network during the applications of an external cue and feedback, whereby the memory pattern is stable only when the feedback exists, and once the feedback is terminated, neural correlations start to interfere the memory pattern again. In this sense, neural feedback in our model does not increase the memory capacity of the network, but rather it provides an efficient way to leverages memory retrieval temporally during the execution of a brain function. Our model suggests a new way to overcome the correlation interference problem in memory retrieval in neural networks.

The present study has focused on exploring how the neural system resolves the 'negative' impact of neural correlations on memory retrieval, but in other computational tasks, neural correlations may have important positive roles. For instance, a modeling study based on the Hopfield model has revealed that if the transition probabilities between memorized patterns are determined by their correlations, then the dynamics of network states naturally reproduce the statistical properties of free call of humans (Recanatesi, Katkov, & Tsodyks, 2017), suggesting that neural correlation plays a crucial role for efficient memory searching in the brain. To fully understand how the brain organizes and manipulates memories, it is necessary to study the effects of neural correlations from alternative aspects.

Finally, in this study, in order to elucidate the underlying computational mechanisms clearly, we have adopted a simple hierarchical Hopfield model which allow us to pursue the theoretical analysis. Although the Hopfield associative memory model is known to lack many biological details, it captures two fundamental properties of neural information processing. First, it captures the property that a neural network utilizes neuronal connections for both information storing and retrieving, realizing in-memory computing. Second, it captures the property that neuronal connections are determined by the representation correlations of objects via Hebbian-like learning rules. Once these two properties hold, the interference of neural correlations to information retrieval becomes inevitable, and our main results about the function of push–pull feedback are applicable. We therefore expect that the present study will give us insight into understanding the general role of feedback in neural information processing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Appendix A. Hierarchical memory patterns

The hierarchical memory patterns used in the present study are constructed as follows Amari and Maginu (1988). First, the grandparent patterns are statistically independent to each other, whose elements have equal probabilities of taking a value of 1 or -1, and are drawn from the distribution,

$$P(\xi_i^\alpha) = \frac{1}{2}\delta(\xi_i^\alpha + 1) + \frac{1}{2}\delta(\xi_i^\alpha - 1), \tag{A.1}$$

where $\delta(x) = 1$ for $x = 0$ and $\delta(x) = 0$ otherwise.

Secondly, for each grandparent pattern, its descending parent patterns are drawn from the distribution,

$$P(\xi_i^{\alpha,\beta}) = (\frac{1+b_2}{2})\delta(\xi_i^{\alpha,\beta} - \xi_i^\alpha) + (\frac{1-b_2}{2})\delta(\xi_i^{\alpha,\beta} + \xi_i^\alpha), \tag{A.2}$$

where $0 < b_2 < 1$, implying that each element of a parent pattern has a probability of $(1 + b_2)/2 > 0.5$ to have the same value as the corresponding element of the grandparent. By this, the descending relationship between a grandparent and its parent patterns is established.

Thirdly, for each parent pattern, its descending child patterns are drawn from the distribution,

$$P(\xi_i^{\alpha,\beta,\gamma}) = (\frac{1+b_1}{2})\delta(\xi_i^{\alpha,\beta,\gamma} - \xi_i^{\alpha,\beta}) + (\frac{1-b_1}{2})\delta(\xi_i^{\alpha,\beta,\gamma} + \xi_i^{\alpha,\beta}), \tag{A.3}$$

where $0 < b_1 < 1$. By this, the descending relationship between a parent and its child patterns is established.

The correlations between the above generated hierarchical memory patterns satisfy the following relationships:

- Between grandparent patterns, $1/N\sum_i \xi_i^\alpha \xi_i^{\alpha'} = 0$, for $\alpha \neq \alpha'$;
- Between grandparent and its descending parent patterns, $1/N\sum_i \xi_i^\alpha \xi_i^{\alpha,\beta} = b_2$;
- Between parent patterns from the same grandparent, $1/N\sum_i \xi_i^{\alpha,\beta} \xi_i^{\alpha,\beta'} = b_2^2$, for $\beta \neq \beta'$;
- Between parent patterns from different grandparents, $1/N\sum_i \xi_i^{\alpha,\beta} \xi_i^{\alpha',\beta'} = 0$, for $\alpha \neq \alpha'$;
- Between parent pattern and its descending child patterns, $1/N\sum_i \xi_i^{\alpha,\beta} \xi_i^{\alpha,\beta,\gamma} = b_1$;
- Between child patterns from the same parent pattern, $1/N\sum_i \xi_i^{\alpha,\beta,\gamma} \xi_i^{\alpha\beta\gamma'} = b_1^2$, for $\gamma \neq \gamma'$;
- Between child patterns from different parents, $1/N\sum_i \xi_i^{\alpha,\beta,\gamma} \xi_i^{\alpha,\beta',\gamma'} = b_2^2 b_1^2$, for $\beta \neq \beta'$.

It can be checked that when the number of parent patterns P_β is sufficiently large, the average of parent patterns approaches to their ascending grandparent pattern, i.e., $1/P_\beta \sum_{\beta=1}^{P_\beta} \xi_i^{\alpha,\beta} \approx b_2 \xi_i^\alpha$. Similarly, when the number of child patterns P_γ is sufficiently large, the average of child patterns approaches to their ascending parent pattern, i.e., $1/P_\gamma \sum_{\gamma=1}^{P_\gamma} \xi_i^{\alpha,\beta,\gamma} \approx b_1 \xi_i^{\alpha,\beta}$.

Appendix B. Theoretical analysis of the network performance

B.1. The neuronal input at layer 1 without feedback

According to Eq. (1) and that $W_{ij}^1 = \sum_{\alpha,\beta,\gamma} \xi_i^{\alpha,\beta,\gamma} \xi_j^{\alpha,\beta,\gamma}/N$, without feedback, we have

$$h_i^1(0) = \frac{1}{N}\sum_j W_{ij}^1 x_j^1(0),$$

$$= \frac{1}{N}\sum_{\alpha,\beta,\gamma}\xi_i^{\alpha,\beta,\gamma}\sum_j \xi_j^{\alpha,\beta,\gamma}\xi_j^{\alpha_0,\beta_0,\gamma_0},$$
$$= \xi_i^{\alpha_0,\beta_0\gamma_0} + A_i + \widetilde{A}_i + \widetilde{\widetilde{A}}_i, \tag{B.1}$$

where

$$A_i = \frac{1}{N}\sum_{\gamma\neq\gamma_0}\xi_i^{\alpha_0,\beta_0,\gamma}\sum_j \xi_j^{\alpha_0,\beta_0,\gamma}\xi_j^{\alpha_0,\beta_0,\gamma_0},$$
$$= b_1^2\sum_{\gamma\neq\gamma_0}\xi_i^{\alpha_0,\beta_0,\gamma}, \tag{B.2}$$

$$\widetilde{A}_i = \frac{1}{N}\sum_{\beta\neq\beta_0}\sum_{\gamma}\xi_i^{\alpha_0,\beta,\gamma}\sum_j \xi_j^{\alpha_0,\beta,\gamma}\xi_j^{\alpha_0,\beta_0,\gamma_0},$$
$$= b_1^2 b_2^2\sum_{\beta\neq\beta_0,\gamma}\xi_i^{\alpha_0,\beta,\gamma}, \tag{B.3}$$

and

$$\widetilde{\widetilde{A}}_i = \frac{1}{N}\sum_{\alpha\neq\alpha_0}\sum_{\beta,\gamma}\xi_i^{\alpha,\beta,\gamma}\sum_j \xi_j^{\alpha,\beta,\gamma}\xi_j^{\alpha_0,\beta_0,\gamma_0},$$
$$= 0. \tag{B.4}$$

According to Eqs. (5) and (B.1), we have

$$C_i = b_1^2\xi_i^{\alpha_0,\beta_0,\gamma_0}\sum_{\gamma\neq\gamma_0}\xi_i^{\alpha_0,\beta_0,\gamma}, \tag{B.5}$$

$$\widetilde{C}_i = b_1^2 b_2^2\xi_i^{\alpha_0,\beta_0,\gamma_0}\sum_{\beta\neq\beta_0}\sum_{\gamma}\xi_i^{\alpha_0,\beta,\gamma}, \tag{B.6}$$

$$\widetilde{\widetilde{C}}_i = 0. \tag{B.7}$$

B.2. The distribution of C_i

Define the sets $\mathcal{C}_+ \equiv \{\gamma \mid \xi_i^{\alpha_0,\beta_0,\gamma} = \xi_i^{\alpha_0,\beta_0}, \gamma \neq \gamma_0\}$, and $\mathcal{C}_- \equiv \{\gamma \mid \xi_i^{\alpha_0,\beta_0,\gamma} = -\xi_i^{\alpha_0,\beta_0}, \gamma \neq \gamma_0\}$, and their numbers are denoted as $card(\mathcal{C}_+)$ and $card(\mathcal{C}_-)$, respectively. Note $card(\mathcal{C}_-) = P_\gamma - 1 - card(\mathcal{C}_+)$.

According to Eq. (A.3),

$$P(\xi_i^{\alpha,\beta,\gamma}) = (\frac{1+b_1}{2})\delta(\xi_i^{\alpha,\beta,\gamma} - \xi_i^{\alpha,\beta}) + (\frac{1-b_1}{2})\delta(\xi_i^{\alpha,\beta,\gamma} + \xi_i^{\alpha,\beta}).$$

Since the generalizations of memory patterns are independent to each other, $card(\mathcal{C}_+)$ follows a binomial distribution, $card(\mathcal{C}_+) \sim \mathcal{B}(P_\gamma - 1, P_+)$, with $P_+ = P(\xi_i^{\alpha_0,\beta_0,\gamma} = \xi_i^{\alpha_0,\beta_0}) = (1+b_1)/2$. Here, $\mathcal{B}(n, P)$ denotes a binomial distribution, with n the total number of experiments and P the probability of each experiment yielding a successful result.

In the large P_γ limit, a binomial distribution can be approximated as a normal one, which is written as

$$card(\mathcal{C}_+) \sim \mathcal{N}\left[(P_\gamma - 1)P_+, (P_\gamma - 1)P_+(1 - P_+)\right]. \tag{B.8}$$

Thus, we have

$$\sum_{\gamma\neq\gamma_0}\xi_i^{\alpha_0,\beta_0,\gamma} = card(\mathcal{C}_+)\xi_i^{\alpha_0,\beta_0} - card(\mathcal{C}_-)\xi_i^{\alpha_0,\beta_0},$$
$$= 2card(\mathcal{C}_+)\xi_i^{\alpha_0,\beta_0} - (P_\gamma - 1)\xi_i^{\alpha_0,\beta_0}, \tag{B.9}$$

and hence

$$C_i = b_1^2\left[2card(\mathcal{C}_+) - (P_\gamma - 1)\right]\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0}. \tag{B.10}$$

Since the product $\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0} = 1$ with the probability $(1 + b_1)/2$ and $\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0} = -1$ with the probability $(1 - b_1)/2$, the distribution of C_i can be written as superposition of two normal distributions, i.e.,

$$P(C_i) = \frac{1+b_1}{2}\mathcal{N}(E_C, V_C) + \frac{1-b_1}{2}\mathcal{N}(-E_C, V_C), \tag{B.11}$$

where $E_C = b_1^3(P_\gamma - 1)$ and $V_C = b_1^4(P_\gamma - 1)(1 - b_1^2)$.

B.3. The distribution of $\widetilde{C}_i$

Similarly, it can be checked that in the large limits of P_γ and P_β, $\widetilde{C}_i$ satisfies superposition of two normal distributions,

$$P(\widetilde{C}_i) = \frac{1+b_1b_2}{2}\mathcal{N}(E_{\widetilde{C}}, V_{\widetilde{C}}) + \frac{1-b_1b_2}{2}\mathcal{N}(-E_{\widetilde{C}}, V_{\widetilde{C}}), \tag{B.12}$$

where $E_{\widetilde{C}} = b_1^3b_2^3P_\gamma(P_\beta - 1)$ and $V_{\widetilde{C}} = b_1^4b_2^4P_\gamma(P_\beta - 1)(1 - b_1^2)(1 - b_2^2)$.

B.4. The distribution of C_i^*

According to the definition, $C_i^* = C_i - b_1\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0}$. It can be checked that for $\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0} = 1$, the mean $E(C_i^*) = E_C - b_1$, and for $\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0} = -1$, the mean $E(C_i^*) = -E_C + b_1$. Therefore, we have

$$P(C_i^*) = \frac{1+b_1}{2}\mathcal{N}(E_C - b_1, V_C^*) + \frac{1-b_1}{2}\mathcal{N}(-E_C + b_1, V_C^*), \tag{B.13}$$

where $E_C = b_1^3(P_\gamma - 1)$ and $V_{C^*} = V_C = b_1^4(P_\gamma - 1)(1 - b_1^2)$.

B.5. The contribution of the pull feedback

We first calculate retrieval error without the pull feedback. The probabilities $P(C_i)$ and $P(\widetilde{C}_i)$ can be calculated under four different conditions, which are summarized below:

- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = \xi_i^{\alpha_0,\beta_0} = \xi_i^{\alpha_0}$, which has the probability $P = (1 + b_1)(1 + b_2)/4$, $P(C_i) = \mathcal{N}(E_C, V_C)$ and $P(\widetilde{C}_i) = \mathcal{N}(E_{\widetilde{C}}, V_{\widetilde{C}})$.
- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = \xi_i^{\alpha_0,\beta_0} = -\xi_i^{\alpha_0}$, which has the probability $P = (1 + b_1)(1 - b_2)/4$, $P(C_i) = \mathcal{N}(E_C, V_C)$ and $P(\widetilde{C}_i) = \mathcal{N}(-E_{\widetilde{C}}, V_{\widetilde{C}})$.
- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = -\xi_i^{\alpha_0,\beta_0} = \xi_i^{\alpha_0}$, which has the probability $P = (1 - b_1)(1 + b_2)/4$, $P(C_i) = \mathcal{N}(-E_C, V_C)$ and $P(\widetilde{C}_i) = \mathcal{N}(-E_{\widetilde{C}}, V_{\widetilde{C}})$.
- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = -\xi_i^{\alpha_0,\beta_0} = -\xi_i^{\alpha_0}$, which has the probability $P = (1 - b_1)(1 - b_2)/4$, $P(C_i) = \mathcal{N}(-E_C, V_C)$ and $P(\widetilde{C}_i) = \mathcal{N}(E_{\widetilde{C}}, V_{\widetilde{C}})$.

The above four conditions can be described by two auxiliary variables, which are $s_1 \equiv \xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0} = \pm 1$ and $s_2 \equiv \xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0} = \pm 1$.

The probability of retrieval error without the pull feedback is calculated to be,

$$p(C_i + \widetilde{C}_i < -1)$$
$$= \int_{-\infty}^{+\infty} p(C_i = x)p(\widetilde{C}_i < -1 - x)dx$$
$$= \sum_{s_1,s_2=\pm1}\frac{1}{4}(1 + s_1b_1)(1 + s_1s_2b_2)$$
$$\times \int_{-\infty}^{+\infty}\frac{dx}{\sqrt{2\pi V_C}}exp(-\frac{x - s_1E_C}{2V_C})\int_{-\infty}^{\frac{-1-x-s_2E_{\widetilde{C}}}{\sqrt{2V_{\widetilde{C}}}}}\frac{dy}{\sqrt{2\pi}}exp(-\frac{y^2}{2}),$$
$$= \sum_{s_1,s_2=\pm1}\frac{1}{4}(1 + s_1b_1)(1 + s_1s_2b_2)$$
$$\times \int_{-\infty}^{+\infty}\frac{dx}{\sqrt{2\pi}}exp(-\frac{x\sigma_c - s_1E_C}{2})\int_{-\infty}^{\frac{-1-x\sigma_{\widetilde{C}}-s_2E_{\widetilde{C}}}{\sigma_{\widetilde{C}}}}\frac{dy}{\sqrt{2\pi}}exp(-\frac{y^2}{2}),$$
$$(\sigma_C = \sqrt{V_c}; \sigma_{\widetilde{C}} = \sqrt{V_{\widetilde{C}}}),$$

$$= \sum_{s_1,s_2=\pm1} \frac{1}{4}(1+s_1b_1)(1+s_1s_2b_2)$$
$$\times \int_{-\infty}^{+\infty} \frac{dx}{\sqrt{2\pi}} exp(-\frac{x^2}{2}) \int_{-\infty}^{+\infty} \frac{dy}{\sqrt{2\pi}} exp(-\frac{y^2}{2}),$$
$$(\sigma_C x + \sigma_{\widetilde{C}} y < -1 - s_1E_C - s_2E_{\widetilde{C}})$$
$$= \sum_{s_1,s_2=\pm1} \frac{1}{8}(1+s_1b_1)(1+s_1s_2b_2)$$
$$\times \left[1 + erf(\frac{-1 - s_1E_C - s_2E_{\widetilde{C}}}{\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}}) \right]. \quad (B.14)$$

We now calculate retrieval error with the pull feedback. Given $W_{ij}^{1,2} = -b_1\delta_{ij}$ and that layer 2 is at the parent pattern $\mathbf{x}^2(0) = \xi^{\alpha_0,\beta_0}$, the alignment between the neuronal input and the child pattern at layer 1 is written as,

$$\xi^{\alpha_0,\beta_0,\gamma_0} h_i^1(0) = 1 + C_i + \widetilde{C}_i - b_1\xi^{\alpha_0,\beta_0}\xi^{\alpha_0,\beta_0,\gamma_0}. \quad (B.15)$$

We obtain

$$m^{\alpha_0,\beta_0,\gamma_0}(1) = \frac{1}{N}\sum_{i=1}^{N} \text{sign}[1 + C_i + \widetilde{C}_i - b_1\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0}],$$
$$= \frac{1}{N}\sum_{i=1}^{N} \text{sign}\left[1 + C_i^* + \widetilde{C}_i\right] \quad (B.16)$$

where $C_i^* = C_i - b_1\xi_i^{\alpha_0,\beta_0,\gamma_0}\xi_i^{\alpha_0,\beta_0}$.

The probability $P(C_i^*)$ can be calculated under four different conditions, which are,

- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = \xi_i^{\alpha_0,\beta_0} = \xi_i^{\alpha_0}$, which has the probability $P = (1+b_1)(1+b_2)/4$, $P(C_i^*) = \mathcal{N}[(E_C - b1), V_C]$;
- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = \xi_i^{\alpha_0,\beta_0} = -\xi_i^{\alpha_0}$, which has the probability $P = (1+b_1)(1-b_2)/4$, $P(C_i^*) = \mathcal{N}[(E_C - b1), V_C]$;
- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = -\xi_i^{\alpha_0,\beta_0} = \xi_i^{\alpha_0}$, which has the probability $P = (1-b_1)(1+b_2)/4$, $P(C_i^*) = \mathcal{N}[-(E_C - b1), V_C]$;
- When $\xi_i^{\alpha_0,\beta_0,\gamma_0} = -\xi_i^{\alpha_0,\beta_0} = -\xi_i^{\alpha_0}$, which has the probability $P = (1-b_1)(1-b_2)/4$, $P(C_i^*) = \mathcal{N}[-(E_C - b1), V_C]$.

Analogy to Eq. (B.14), the probability of retrieval error with the pull feedback is calculated to be,

$$p(C_i^* + \widetilde{C}_i < -1)$$
$$= \sum_{s_1,s_2=\pm1} \frac{(1+s_1b_1)(1+s_1s_2b_2)}{8}$$
$$\times \left\{ 1 + erf\left[\frac{-1 - s_1(E_C - b_1) - s_2E_{\widetilde{C}}}{\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}} \right] \right\}. \quad (B.17)$$

B.6. The pull feedback improves the retrieval accuracy

We approximate the error function with the 3rd order Taylor expansion, i.e.,

$$erf(z) \approx \frac{2}{\sqrt{\pi}}\left(z - \frac{z^3}{3}\right).$$

The change of the error probability after applying the pull feedback is calculated to be,

$$p(C_i + \widetilde{C}_i < -1) - p(C_i^* + \widetilde{C}_i < -1)$$
$$\approx \sum_{s_1,s_2=\pm1} \frac{(1+s_1b_1)(1+s_1s_2b_2)}{8}$$
$$\times \left\{ \frac{s_1b_1}{3\left[\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}\right]^3} \left(3z^2 - 3s_1b_1z + b_1^2\right) \right\},$$
$$(z = s_1E_C + s_2E_{\widetilde{C}},)$$
$$= \sum_{s_1,s_2=\pm1} \frac{(1+s_1b_1)(1+s_1s_2b_2)}{4\sqrt{\pi}}$$
$$\frac{1}{\left[\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}\right]^3}\left[3s_1b_1z^2 - 3b_1^2z + s_1b_1^3\right],$$
$$= \frac{1}{3\sqrt{\pi}\left[\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}\right]^3}$$
$$\times \left[3b_1^7(P_\gamma^2 - b_1) - 9b_1^6(P_\gamma - 1) + b_1^4(6P_\gamma - 1)\right],$$
$$= \frac{b_1^4}{3\sqrt{\pi}\left[\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}\right]^3}$$
$$\times \left[3b_1^3(P_\gamma^2 - b_1) - 9b_1^2(P_\gamma - 1) + (6P_\gamma - 1)\right]. \quad (B.18)$$

Under the condition $P_\gamma \gg 1$, we have

$$p(C_i + \widetilde{C}_i < -1) - p(C_i^* + \widetilde{C}_i < -1)$$
$$\approx \frac{1}{3\sqrt{\pi}\left[\sqrt{2(\sigma_C^2+\sigma_{\widetilde{C}}^2)}\right]^3}\left(3b_1^7P_\gamma^2 - 9b_1^6P_\gamma + 6b_1^4P_\gamma\right)$$
$$> 0. \quad (B.19)$$

To get the above result, we used the condition that $3b_1^7P_\gamma^2 - 9b_1^6P_\gamma + 6b_1^4P_\gamma > 0$ for $P_\gamma > 2$. Thus, in general cases of $P_\gamma \gg 1$, which is true in reality, the pull feedback guarantees to improve the retrieval accuracy of the child pattern.

B.7. The pull feedback de-correlates sibling patterns

The average correlation between sibling patterns of the same parent is given by, $CR = 1/[N(P_\gamma - 1)]\sum_{i=1}^{N}\sum_{\gamma'\neq\gamma}\xi_i^{\alpha^0,\beta^0,\gamma^0}\xi_i^{\alpha^0,\beta^0,\gamma'} = b_1^2$.

With the pull feedback input, $-b_1\xi^{\alpha^0,\beta^0}$, the average correlation between sibling patterns becomes,

$$CR = \frac{1}{N(P_\gamma - 1)}\sum_{i=1}^{N}\sum_{\gamma'\neq\gamma}\left(\xi_i^{\alpha^0,\beta^0,\gamma^0} - b_1\xi_i^{\alpha^0,\beta^0}\right)$$
$$\times \left(\xi_i^{\alpha^0,\beta^0,\gamma'} - b_1\xi_i^{\alpha^0,\beta^0}\right)$$
$$= \frac{1}{N(P_\gamma - 1)}$$
$$\sum_{i=1}^{N}\sum_{\gamma'\neq\gamma}\left(\xi_i^{\alpha^0,\beta^0,\gamma^0} - b_1\xi_i^{\alpha^0,\beta^0,\gamma^0}\xi_i^{\alpha^0,\beta^0}\right.$$
$$\left. - b_1\xi_i^{\alpha^0,\beta^0,\gamma'}\xi_i^{\alpha^0,\beta^0} + b_1^2\right)$$
$$= \frac{1}{P_\gamma - 1}$$
$$\sum_{\gamma'\neq\gamma}\left[\frac{(1+b_1)^2}{4}(1-2b_1) + \frac{(1-b_1)^2}{4}(1+2b_1)\right.$$
$$\left. - \frac{1-b_1^2}{2} + b_1^2\right]$$
$$= 0. \quad (B.20)$$

Appendix C. Simulation on the continuous Hopfield model

C.1. Dynamics of the two-layer network

The dynamics of the continuous Hopfield model in the two-layer network are given by

$$\tau \frac{dh_i^1}{dt} = -h_i^1 + \sum_j W_{ij}^1 x_j^1 + \sum_k W_{ik}^{1,2}(t) x_k^2 + I_i^{ext,1}, \tag{C.1}$$

$$\tau \frac{dh_i^2}{dt} = -h_i^2 + \sum_j W_{ij}^2 x_j^2 + \sum_k W_{ik}^{2,1}(t) x_k^1 + I_i^{ext,2}, \tag{C.2}$$

$$x_i^n = \frac{1}{\pi} \arctan(8\pi h_i^n) + 0.5, \quad \text{for} \quad n = 1, 2, \tag{C.3}$$

where h_i^n and x_i^n denote the synaptic input and the firing rate of neuron i in layer n. τ is the time constant, and $arctan(x)$ is the inverse of the tangent function. $I_i^{ext,n}$ the external input to layer n, and the difference between $I_i^{ext,1}$ and $I_i^{ext,2}$ is only in the strength.

In the continuous model, each element of a memory pattern (e.g., $\xi_i^{\alpha,\beta,\gamma}$) takes a value of 0 or 1. The recurrent connections between neurons in the same layer are constructed by the Hebbian covariance learning rule, which are given by

$$\tilde{W}_{ij}^1 = \frac{1}{N} \sum_{\alpha,\beta,\gamma} \left(\xi_i^{\alpha,\beta,\gamma} - \langle \xi \rangle_{\alpha,\beta,\gamma} \right) \left(\xi_j^{\alpha,\beta,\gamma} - \langle \xi \rangle_{\alpha,\beta,\gamma} \right), \tag{C.4}$$

$$\tilde{W}_{ij}^2 = \frac{1}{N} \sum_{\alpha,\beta} \left(\xi_i^{\alpha,\beta} - \langle \xi \rangle_{\alpha,\beta} \right) \left(\xi_j^{\alpha,\beta} - \langle \xi \rangle_{\alpha,\beta} \right), \tag{C.5}$$

$$W_{ij}^n = a_r^n \frac{\tilde{W}_{ij}^n}{|\tilde{\mathbf{W}}^n|}, \quad \text{for} \quad n = 1, 2, \tag{C.6}$$

where $\langle \xi \rangle_*$ is the mean activity of all neurons averaged over all memory patterns in the layer $*$, $|\tilde{\mathbf{W}}^n| = \sqrt{\sum_{ij} (\tilde{W}_{ij}^n)^2 / N^2}$, and a_r^n are positive constants.

Similar to the recurrent connections, the feedforward connections from low layer to high layer are given by

$$\tilde{W}_{ik}^{2,1} = \frac{1}{N} \sum_{\alpha,\beta,\gamma} \left(\xi_i^{\alpha,\beta} - \langle \xi \rangle_{\alpha,\beta} \right) \left(\xi_k^{\alpha,\beta,\gamma} - \langle \xi \rangle_{\alpha,\beta,\gamma} \right), \tag{C.7}$$

$$W_{ik}^{2,1} = a_1^2 \frac{\tilde{W}_{ik}^{2,1}}{|\tilde{\mathbf{W}}^{2,1}|}, \tag{C.8}$$

where $\langle \xi \rangle_*$ is the mean activity of all neurons averaged over all memory patterns in the layer $*$, $|\tilde{\mathbf{W}}^{2,1}| = \sqrt{\sum_{ik} (\tilde{W}_{ik}^{2,1})^2 / N^2}$, and a_1^2 are positive constants.

We consider the feedback connections from layer 2 to 1, which vary over time. At the early phase, the push feedback is applied, which is written as

$$W_{ik}^{1,2} = \frac{a_+}{NP_\gamma} \sum_{\alpha,\beta,\gamma} (\xi_i^{\alpha,\beta,\gamma} - \langle \xi \rangle_{\alpha,\beta,\gamma})(\xi_k^{\alpha,\beta} - \langle \xi \rangle_{\alpha,\beta}), \tag{C.9}$$

where a_+ is a positive number.

At the later phase, the pull feedback is used which is written

$$W_{ik}^{1,2} = -a_- b_1 \delta_{ik}, \tag{C.10}$$

where $\delta_{ik} = 1$ for $i = k$ and $\delta_{ik} = 0$ otherwise, and a_- is a positive number.

Let us consider the memory pattern to be retrieved at layer 1 is $\xi^{\alpha_0,\beta_0,\gamma_0}$. The external input to the first and top layers, which conveys the memory pattern information, is set to be

$$I_i^{ext,n} = a_{ext}^n (\tilde{\xi}_i^{\alpha_0,\beta_0,\gamma_0} + \sigma \eta_i), \quad \text{for} \quad n = 1, 2, \tag{C.11}$$

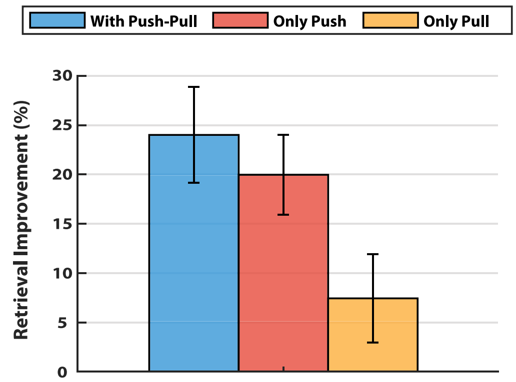

Fig. C.8. Comparing network performances with the push–pull, only the push, and only the pull feedbacks, respectively. With $P_\beta = 2$, $P_\gamma = 20$, $b_1 = 0.3$, $\lambda_1 = 0.4$, other parameters are the same as in Fig. 4 in the main text.

where η_i is a random number uniformly distributed in the range of $(-1, 1)$, representing the memory-independent noise, and σ controls the noise strength. a_{ext} is a positive number.

The pattern $\tilde{\xi}^{\alpha_0,\beta_0,\gamma_0}$ is the noise-corrupted signal, which is constructed as follows: starting from the clean memory pattern $\xi^{\alpha_0,\beta_0,\gamma_0}$, we first randomly select $\lambda_1 N$ number of neurons, with $0 < \lambda_1 < 1$, and change their values to match a sibling pattern $\xi^{\alpha_0,\beta_0,\gamma'}$, with $\gamma' \neq \gamma_0$, which generates the intra-class noise; we then randomly select $\lambda_2 N$ number of neurons, with $0 < \lambda_2 < 1$, and change their values to match a cousin pattern $\xi^{\alpha_0,\beta',\gamma_0}$, with $\beta' \neq \beta_0$, which generates the inter-class noise.

C.2. The effect of push feedback with different parameters

According to Fig. 5, it may appear that the pull feedback is completely dominating, while the effect of push feedback is negligible. Actually, this varies with the parameters. Below is an example, which shows that applying only push actually outperforms applying only pull, and when they are applied in the right temporal order, the network has the best performance. In general, we find that when the number of patterns is fewer (or the duration of the external input is short), the push feedback tends to have a larger contribution.

C.3. Dynamics of the three-layer network

Similar to the two-layer network in Appendix C.1, the dynamics of the three-layer network as shown in Fig. 6 are given by

$$\tau \frac{dh_i^1}{dt} = -h_i^1 + \sum_j W_{ij}^1 x_j^1 + \sum_k W_{ik}^{1,2}(t) x_k^2 + I_i^{ext,1}, \tag{C.12}$$

$$\tau \frac{dh_i^2}{dt} = -h_i^2 + \sum_j W_{ij}^2 x_j^2 + \sum_k W_{ik}^{2,1}(t) x_k^1 + \sum_k W_{ik}^{2,3}(t) x_k^3, \tag{C.13}$$

$$\tau \frac{dh_i^3}{dt} = -h_i^3 + \sum_j W_{ij}^3 x_j^3 + \sum_k W_{ik}^{3,2}(t) x_k^2 + I_i^{ext,3}, \tag{C.14}$$

$$x_i^n = \frac{1}{\pi} \arctan(8\pi h_i^n) + 0.5, \quad \text{for} \quad n = 1, 2, 3, \tag{C.15}$$

where h_i^n and x_i^n denote the synaptic input and the firing rate of neuron i in layer n, respectively, τ the time constant, $I_i^{ext,n}$ the external input to layer n, $n = 1$ or 3.

The recurrent connections between neurons in the same layers are given by:

$$\tilde{W}_{ij}^{1} = \frac{1}{N}\sum_{\alpha,\beta,\gamma}\left(\xi_i^{\alpha,\beta,\gamma} - \langle\xi\rangle_{\alpha,\beta,\gamma}\right)\left(\xi_j^{\alpha,\beta,\gamma} - \langle\xi\rangle_{\alpha,\beta,\gamma}\right), \tag{C.16}$$

$$\tilde{W}_{ij}^{2} = \frac{1}{N}\sum_{\alpha,\beta}\left(\xi_i^{\alpha,\beta} - \langle\xi\rangle_{\alpha,\beta}\right)\left(\xi_j^{\alpha,\beta} - \langle\xi\rangle_{\alpha,\beta}\right), \tag{C.17}$$

$$\tilde{W}_{ij}^{3} = \frac{1}{N}\sum_{\alpha}\left(\xi_i^{\alpha} - \langle\xi\rangle_{\alpha}\right)\left(\xi_j^{\alpha} - \langle\xi\rangle_{\alpha}\right), \tag{C.18}$$

$$W_{ij}^{n} = a_r^n \frac{\tilde{W}_{ij}^{n}}{|\tilde{\mathbf{W}}^n|}, \quad \text{for} \quad n = 1, 2, 3 \tag{C.19}$$

where $\langle\xi\rangle_*$ is the mean activity of all neurons averaged over all memory patterns in the layer $*$, $|\tilde{\mathbf{W}}^n| = \sqrt{\sum_{ij}(\tilde{W}_{ij}^n)^2/N^2}$, and a_r^n are positive constants.

Similar to the recurrent connections, the feedforward connections from low layer to high layer are given by

$$\tilde{W}_{ik}^{2,1} = \frac{1}{N}\sum_{\alpha,\beta,\gamma}\left(\xi_i^{\alpha,\beta} - \langle\xi\rangle_{\alpha,\beta}\right)\left(\xi_k^{\alpha,\beta,\gamma} - \langle\xi\rangle_{\alpha,\beta,\gamma}\right), \tag{C.20}$$

$$\tilde{W}_{ik}^{3,2} = \frac{1}{N}\sum_{\alpha,\beta}\left(\xi_i^{\alpha} - \langle\xi\rangle_{\alpha}\right)\left(\xi_k^{\alpha,\beta} - \langle\xi\rangle_{\alpha,\beta}\right), \tag{C.21}$$

$$W_{ik}^{n+1,n} = a_n^{n+1}\frac{\tilde{W}_{ik}^{n+1,n}}{|\mathbf{W}^{\widetilde{n+1,n}}|}, \quad \text{for} \quad n = 1, 2, \tag{C.22}$$

where $\langle\xi\rangle_*$ is the mean activity of all neurons averaged over all memory patterns in the layer $*$, $|\tilde{\mathbf{W}}^{n+1,n}| = \sqrt{\sum_{ik}(\tilde{W}_{ik}^{n+1,n})^2/N^2}$, and a_n^{n+1} are positive constants.

At the early phase, the push feedback connections from high to low layer can be written as:

$$W_{ik}^{1,2} = \frac{a_+^1}{NP_\gamma}\sum_{\alpha,\beta,\gamma}(\xi_i^{\alpha,\beta,\gamma} - \langle\xi\rangle_{\alpha,\beta,\gamma})(\xi_k^{\alpha,\beta} - \langle\xi\rangle_{\alpha,\beta}), \tag{C.23}$$

$$W_{ik}^{2,3} = \frac{a_+^2}{NP_\gamma}\sum_{\alpha,\beta}(\xi_i^{\alpha,\beta} - \langle\xi\rangle_{\alpha,\beta})(\xi_k^{\alpha} - \langle\xi\rangle_{\alpha}), \tag{C.24}$$

where a_+^1 and a_+^2 are positive numbers.

At the later phase, the push feedback connections are written as:

$$W_{ik}^{1,2} = -a_-^1 b_1 \delta_{ik}, \tag{C.25}$$

$$W_{ik}^{2,3} = -a_-^2 b_2 \delta_{ik}, \tag{C.26}$$

where $\delta_{ik} = 1$ for $i = k$ and $\delta_{ik} = 0$ otherwise, and a_-^1, a_-^2 are positive numbers.

The external input to the first and top layers is set to be:

$$I_i^{ext,n} = a_{ext}^n(\tilde{\xi}_i^{\alpha_0,\beta_0,\gamma_0} + \sigma\eta_i), \quad \text{for} \quad n = 1, 3, \tag{C.27}$$

where η_i is a random number uniformly distributed in the range of $(-1, 1)$, representing the memory-independent noise, and σ controls the noise strength. a_{ext} is a positive number.

Appendix D. Pre-processing real images with a deep neural network

The dataset we use, chosen from ImageNet, consists of three types (bird, cat and dog), and each type is made up of 4 sub-types. They are: (1) sub-types of bird: Cock, Eagle, Parrot, Ptarmigan; (2) sub-types of cat: Abyssinian Cat, Angora Cat, Burmese Cat, Egyptian Cat; (3) sub-types of dog: Border terrier, Cairn terrier, Pug Dog, Saint Bernard. We chose 100 images per sub-type. We re-scaled each image, with its width and height to be (256, 256), and pixel values in the range of [0, 255]. After that, we presented each image as an input to the VGG16 (Simonyan & Zisserman, 2014) (the VGG16 itself was pre-trained by ImageNet) and took the neural activity before the reading-out layer as the neural representation of the image, referred to as $\hat{x}$ hereafter. Subsequently, we normalized neural activities, i.e.,

$$\widetilde{x}_i = \log(1 + \hat{x}_i), \tag{D.1}$$

$$x_i = \mathrm{ReLu}(\frac{\widetilde{x}_i - \langle\widetilde{x}_i\rangle}{\sigma_{\widetilde{x}_i}}), \tag{D.2}$$

where $\mathrm{ReLU}(\cdot) = \max(0, \cdot)$ is the rectified linear function, and $\langle\widetilde{x}\rangle$ and $\sigma_{\widetilde{x}}$ are the mean and standard deviation of neural activities over all images.

Finally, we generated memory patterns for the hierarchical network. For a child pattern, it is given by

$$\xi_i^{\beta_0,\gamma_0} = (\mathrm{sign}(\langle x_i\rangle_{\beta_0,\gamma_0}) + 1)/2, \tag{D.3}$$

where the average is over all the images belonging to the same sub-type of animal.

The corresponding parent patterns are constructed to be

$$\xi_i^{\beta_0} = (\mathrm{sign}(\langle\xi_i^{\beta_0,\gamma}\rangle_\gamma) + 1)/2, \tag{D.4}$$

where the average is over all child patterns belonging to the same parent.

Appendix E. Simulation protocols

Fig. 2

The discrete network dynamics as given in Eqs. (1)–(2) are used. The parameters are $N = 1000$, $P_\alpha = 2$, $P_\beta = 3$, $P_\gamma = 15$, $b_1 = 0.15$, $b_2 = 0.1$, $\lambda_1 = 0.2$ and $\lambda_2 = 0.1$. The memory patterns are generated according to Eqs. (A.1)–(A.3). In Fig. 2B, the distributions of the intra-class interference C_i and the inter-class interference $\widetilde{C}_i$ are obtained by averaging over 10000 trials using the theoretical results of Eqs. (B.10)–(B.11).

Fig. 3

The same network structure and parameters as in Fig. 2 are used. In Fig. 3A&B, the target child pattern is $\xi^{1,1,1}$, and layer 2 is fixed to be at the parent pattern $\xi^{1,1}$. The retrieval accuracy $m^{1,1,1}$ is calculated by Eq. (4). In Fig. 3B, the mean retrieval accuracies of sibling and cousin patterns are given by $\sum_{i\neq 1} m^{1,1,i}/(P_\gamma - 1)$ and $\sum_i m^{1,2,i}/P_\gamma$, respectively, and they are measured at the moment of 40 simultaneous updates and obtained by averaging over 100 trials. In Fig. 3C, the distribution of the new noise term C_i^* are obtained by averaging over 10 000 trials using the theoretical result of Eq. (B.13).

Fig. 4

The continuous Hopfield model as introduced in Appendix C.1 is used. The parameters are: $N = 2000$, $P_\gamma = 25$, $P_\beta = 4$, $P_\alpha = 2$, $b_1 = 0.2$, $b_2 = 0.1$, $\tau = 5$, $a_{ext}^1 = 1$, $a_{ext}^2 = 0.1$, $a_r^1 = 1$, $a_r^2 = 2$, $a_1^2 = 2$, $a_+ = 1$, $a_- = 8$, $\delta t = 0.02\tau$, $\lambda_1 = 0.1$, $\lambda_2 = 0.1$ and $\sigma = 0.05$. The times for applying the external input, the push feedback, and the pull feedback are $(0-6)\tau$, $(1-2)\tau$, and $(2-3)\tau$, respectively. In Fig. 4A, the mean neural activity $\langle x\rangle$ is obtained by averaging over all neurons in layer 1. In Fig. 4B, the retrieval accuracy is calculated by Eq. (4). In Fig. 4C–F, the retrieval improvement is calculated by $(m_{with-fb} - m_{without-fb}/m_{without-fb})$, with $(m_{with-fb})$ and $m_{without-fb}$

representing the retrieval accuracies with and without feedback, respectively. The statistical results are obtained at the moment of 5τ by averaging over 50 trails, and error bars indicate the standard error of the mean (SEM).

Fig. 5

The same network model and parameters are used as in Fig. 4. The statistical results are obtained at the moment of 5τ by averaging over 50 trails.

Fig. 6

The dynamics of the three-layer network as shown in Fig. 6 are introduced in Appendix C.3. The parameters are: $N = 1000$, $P_\gamma = 25$, $P_\beta = 4$, $P_\alpha = 2$, $b_1 = 0.2$, $b_2 = 0.2$, $\tau = 5$, $a^1_{ext} = 1$, $a^3_{ext} = 0.1$, $a^1_r = 1$, $a^2_r = 2$, $a^3_r = 3$, $a^2_1 = 6$, $a^3_2 = 6$, $a^1_+ = 6$, $a^2_+ = 2$, $a^1_- = 5$, $a^2_- = 7$, $\delta t = 0.02\tau$, $\lambda_1 = 0.15$, $\lambda_2 = 0.15$ and $\sigma = 0.05$. Two rounds of push–pull feedback are applied. Each round lasts for 2τ, with the push and pull feedbacks lasting for 1τ. The feedback starts at 0.2τ. The results are measured at the moment of 5τ and averaged over 100 trials.

Fig. 7

The two-layer network model is used. The parameters are: $N = 4096$, $a^1_r = 1$, $a^2_r = 2$, $a^1_{ext} = 6$, $a^2_{ext} = 1$, $a_+ = 2$, $a_- = 1.5$, $a^2_1 = 1$. The time for applying the external input is $(0 - 5)\tau$, for the push feedback is $(1 - 2.4)\tau$, and for the pull feedback is $(2.4-3.8)\tau$. Other parameters are the same as in Fig. 4. In Fig. 7E, the novel images (cars) are first pre-processed as described in Appendix D to get the corresponding neural representations. The mean network responses for the novel and familiar (birds, cats and dogs) objects are obtained by averaging over 20 randomly chosen images.

References

Ahissar, Merav, & Hochstein, Shaul (2004). The reverse hierarchy theory of visual perceptual learning. *Trends in Cognitive Sciences*, *8*(10), 457–464.

Amari, Shun-Ichi (1972). Learning patterns and pattern sequences by self-organizing nets of threshold elements. *IEEE Transactions on Computers*, *100*(11), 1197–1206.

Amari, Shun-Ichi, & Maginu, Kenjiro (1988). Statistical neurodynamics of associative memory. *Neural Networks*, *1*(1), 63–73.

Carlson, Thomas A., Simmons, Ryan A., Kriegeskorte, Nikolaus, & Slevc, L. Robert (2014). The emergence of semantic meaning in the ventral temporal pathway. *Journal of Cognitive Neuroscience*, *26*(1), 120–131.

Chen, Lin (1982). Topological structure in visual perception. *Science*, *218*(4573), 699–700.

Chen, Lin (2005). The topological approach to perceptual organization. *Visual Cognition*, *12*(4), 553–637.

Chen, Rujia, Wang, Feng, Liang, Hualou, & Li, Wu (2017). Synergistic processing of visual contours across cortical layers in V1 and V2. *Neuron*, *96*(6), 1388–1402.

Chen, Minggui, Yan, Yin, Gong, Xiajing, Gilbert, Charles D., Liang, Hualou, & Li, Wu (2014). Incremental integration of global contours through interplay between visual cortical areas. *Neuron*, *82*(3), 682–694.

Connolly, Andrew C., Guntupalli, J. Swaroop, Gors, Jason, Hanke, Michael, Halchenko, Yaroslav O., Wu, Yu-Chien, et al. (2012). The representation of biological classes in the human brain. *Journal of Neuroscience*, *32*(8), 2608–2618.

Cortes, C., Krogh, A., & Hertz, J. A. (1987). Hierarchical associative networks. *Journal of Physics A: Mathematical and General*, *20*(13), 4449–4455.

Edelman, Shimon (1998). Representation is representation of similarities. *Behavioral and Brain Sciences*, *21*(4), 449–467.

Engel, A. (1990). Storage of hierarchically correlated patterns. *Journal of Physics A: Mathematical and General*, *23*(12), 2587.

Frangou, Polytimi, Correia, Marta, & Kourtzi, Zoe (2018). GABA, not BOLD, reveals dissociable learning-dependent plasticity mechanisms in the human brain. *eLife*, *7*, Article e35854.

Gilad, Ariel, Meirovithz, Elhanan, & Slovin, Hamutal (2013). Population responses to contour integration: early encoding of discrete elements and late perceptual grouping. *Neuron*, *78*(2), 389–402.

Gilbert, Charles D., & Li, Wu (2013). Top-down influences on visual processing. *Nature Reviews Neuroscience*, *14*(5), 350–363.

Hertz, John, Krogh, Anders, & Palmer, Richard G. (2018). *Introduction to the theory of neural computation*. CRC Press.

Hopfield, John J. (1982). Neural networks and physical systems with emergent collective computational abilities. *Proceedings of the National Academy of Sciences*, *79*(8), 2554–2558.

Hopfield, John J. (1984). Neurons with graded response have collective computational properties like those of two-state neurons. *Proceedings of the National Academy of Sciences*, *81*(10), 3088–3092.

Kandel, Eric R., Schwartz, James H., Jessell, Thomas M., & Department of Biochemistry and Molecular Biophysics Thomas Jessell and Siegelbaum, Steven and Hudspeth, A. J. (2000). *Principles of neural science, Vol. 4*. New York: McGraw-hill.

Klink, P. Christiaan, Dagnino, Bruno, Gariel-Mathis, Marie-Alice, & Roelfsema, Pieter R. (2017). Distinct feedforward and feedback effects of microstimulation in visual cortex reveal neural mechanisms of texture segregation. *Neuron*, *95*(1), 209–220.

Kriegeskorte, Nikolaus, & Kievit, Rogier A. (2013). Representational geometry: integrating cognition, computation, and the brain. *Trends in Cognitive Sciences*, *17*(8), 401–412.

Kriegeskorte, Nikolaus, Mur, Marieke, Ruff, Douglas A., Kiani, Roozbeh, Bodurka, Jerzy, Esteky, Hossein, et al. (2008). Matching categorical object representations in inferior temporal cortex of man and monkey. *Neuron*, *60*(6), 1126–1141.

Kropff, Emilio, & Treves, Alessandro (2007). Uninformative memories will prevail: the storage of correlated representations and its consequences. *HFSP Journal*, *1*(4), 249–262.

LeCun, Yann, Bengio, Yoshua, & Hinton, Geoffrey (2015). Deep learning. *Nature*, *521*(7553), 436–444.

Lee, Tai Sing, & Mumford, David (2003). Hierarchical Bayesian inference in the visual cortex. *Journal of the Optical Society of America A*, *20*(7), 1434–1448.

Li, Ya, Wang, Yonghui, & Li, Sheng (2019). Recurrent processing of contour integration in the human visual cortex as revealed by fMRI-guided TMS. *Cerebral Cortex*, *29*(1), 17–26.

Liu, Ling, Wang, Fan, Zhou, Ke, Ding, Nai, & Luo, Huan (2017). Perceptual integration rapidly activates dorsal visual pathway to guide local processing in early visual areas. *PLoS Biology*, *15*(11), Article e2003646.

Liu, Xiao, Zou, Xiaolong, Ji, Zilong, Tian, Gengshuo, Mi, Yuanyuan, Huang, Tiejun, et al. (2019). Push-pull feedback implements hierarchical information retrieval efficiently. In H. Wallach, H. Larochelle, A. Beygelzimer, F. d'Alché Buc, E. Fox, R. Garnett (Eds.), *Advances in neural information processing systems, Vol. 32*. Curran Associates, Inc..

Matsumoto, Narihisa, Okada, Masato, Sugase-Miyamoto, Yasuko, & Yamane, Shigeru (2005). Neuronal mechanisms encoding global-to-fine information in inferior-temporal cortex. *Journal of Computational Neuroscience*, *18*(1), 85–103.

Mur, Marieke, Meys, Mirjam, Bodurka, Jerzy, Goebel, Rainer, Bandettini, Peter A., & Kriegeskorte, Nikolaus (2013). Human object-similarity judgments reflect and transcend the primate-IT object representation. *Frontiers in Psychology*, *4*, 128.

Okada, Masato (2006). Part 3: Brain science, information science and associative memory model. *New Generation Computing*, *24*(2), 185.

Parga, Nestor, & Virasoro, Miguel Angel (1986). The ultrametric organization of memories in a neural network. *Journal de Physique*, *47*(11), 1857–1864.

Perez-Vicente, C. J. (1989). Hierarchical neural network with high storage capacity. *Physical Review A*, *40*(9), 5356.

Rao, Rajesh P. N., & Ballard, Dana H. (1999). Predictive coding in the visual cortex: a functional interpretation of some extra-classical receptive-field effects. *Nature Neuroscience*, *2*(1), 79–87.

Recanatesi, Stefano, Katkov, Mikhail, & Tsodyks, Misha (2017). Memory states and transitions between them in attractor neural networks. *Neural Computation*, *29*(10), 2684–2711.

Sillito, Adam M., Cudeiro, Javier, & Jones, Helen E. (2006). Always returning: feedback and sensory processing in visual cortex and thalamus. *Trends in Neurosciences*, *29*(6), 307–316.

Simonyan, Karen, & Zisserman, Andrew (2014). Very deep convolutional networks for large-scale image recognition. arXiv preprint arXiv:1409.1556.

Sourlas, N. (1988). Multilayer neural networks for hierarchical patterns. *EPL (Europhysics Letters)*, *7*(8), 749–753.

Sugase, Yasuko, Yamane, Shigeru, Ueno, Shoogo, & Kawano, Kenji (1999). Global and fine information coded by single neurons in the temporal visual cortex. *Nature*, *400*(6747), 869–873.

Tsodyks, M. V. (1990). Hierarchical associative memory in neural networks with low activity level. *Modern Physics Letters B*, *4*(04), 259–265.

Wu, Si, Wong, K. Y. Michael, Fung, C. C. Alan, Mi, Yuanyuan, & Zhang, Wenhao (2016). Continuous attractor neural networks: candidate of a canonical model for neural information representation. *F1000Research*, *5*.

Xia, Qiangfei, & Yang, J. Joshua (2019). Memristive crossbar arrays for brain-inspired computing. *Nature Materials*, *18*(4), 309–323.

Yamins, Daniel L. K., Hong, Ha, Cadieu, Charles F., Solomon, Ethan A., Seibert, Darren, & DiCarlo, James J. (2014). Performance-optimized hierarchical models predict neural responses in higher visual cortex. *Proceedings of the National Academy of Sciences*, *111*(23), 8619–8624.

Zou, Xiaolong, Ji, Zilong, Liu, Xiao, Mi, Yuanyuan, Wong, K. Y. Michael, & Wu, Si (2017). Learning a continuous attractor neural network from real images. In Derong Liu, Shengli Xie, Yuanqing Li, Dongbin Zhao, & El-Sayed M. El-Alfy (Eds.), *Neural information processing* (pp. 622–631). Cham: Springer International Publishing.

Zucker, Robert S., & Regehr, Wade G. (2002). Short-term synaptic plasticity. *Annual Review of Physiology*, *64*(1), 355–405.